



1. Executive Summary This design report details Texas A&M University's (TAMU) complete design, manufacturing, and testing of REV Force 2, the team's RC aircraft. Based on the mission requirements, the low-wing conventional tube and wing is made to carry a large amount of passengers and payload, and deploy a banner. The aircraft must complete 3 unique missions. M1 is proof of flight mission that is required to attempt other missions. M2 is a charter flight with passengers and cargo. M3 is a banner towing flight requiring safe deployment and release of the banner. Lastly, the GM is a timed mission which requires a member of the team to complete all attachments on the aircraft.

The TAMU DBF team designed two iterations of aircraft throughout the year. The process was broken into 3 phases: conceptual, preliminary, and detailed design. The conceptual phase involved defining subsystem design requirements, performing a sensitivity study, and several trade studies. For preliminary design, individual sizing was done for the wings, fuselage, and empennage. Mission payloads and the banner were also sized. The detailed design was to finalize the sizing parameters for the aircraft, along with subsystems and propulsion. After the design was completed, a manufacturing and testing timeline was constructed in order to validate the design of the aircraft. All tests were compiled to evaluate the performance of the aircraft

As shown in figure \ref{fig:REV Force 2}, REV Force 2 is a low wing design, with a single tractor motor, a long fuselage to score a large payload, and a standard empanage. The propulsion system was optimized for battery efficiency while still maintaining a long flight time. A performance prediction for the aircraft is shown in Table \ref{table:Predicted Preliminary Performance}

Predicted Preliminary Performance				
Parameter	M1	M2	M3	GM
Aircraft Weight (lbs)	15.93	19.85	16.48	-
Payload Weight (lbs)	-	4.95	0.19	5.14
Banner Length (ft)	-	-	5.83	-
Cruise Speed (ft/s)	85	85	75	-
Turn Speed (ft/s)	50	50	50	-
Lap Time (s/lap)	60	60	75	-
Number of Laps	3	5	4	-
Mission Time (sec)	180	300	300	62

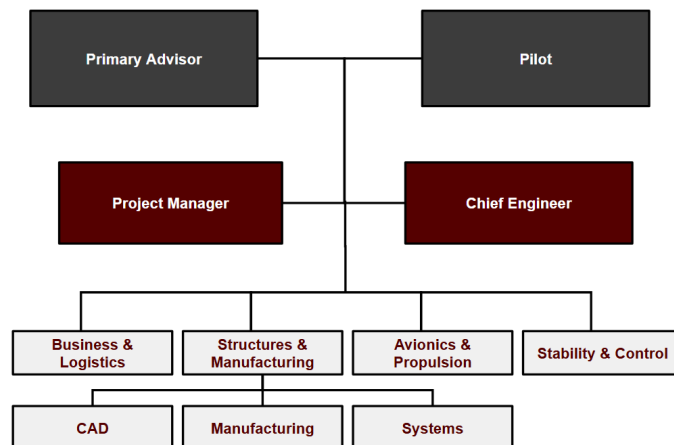


2. Management Summary

2.1 Team Organization and Roles

The Texas A&M DBF team is a student-led design team with 41 student members, two faculty members, and one graduate student advisor. The team is led by eight student members, two overseeing managerial tasks and six overseeing technical tasks as subteam leads. The organization of the leadership team is designed to encourage communication between subteam leads and management. Each subteam lead guides a group of members with their own distinct focus. Figure X is a visual representation of the organizational structure, with the Chief Engineer and Project Manager directly overseeing the subteam leads. The Chief Engineer ensures information is consistent between all technical leads and reviews all major design choices for the aircraft, while the Project Manager maintains deadlines for the entire team, organizes meetings, and manages external communication. All subteams are led by a singular lead, except for the Structures & Manufacturing subteam, which is run by three co-leads: CAD, Manufacturing, and Systems. The responsibilities and required skill sets of each subteam are listed in Table X.

Subteam Skills and Responsibilities		
Technical Lead	Responsibility	Skills Sets
Stability & Control	Scoring Analysis, Aircraft Sizing, CFD Simulations	Thorough knowledge in aerodynamic principles and optimization techniques
Avionics & Propulsion	Power Train Configuration, Test Sensors, Landing Gear	Understanding of electronics and propulsion systems
Business & Logistics	Oversee Finance, Plan Events, Technical Writing and Formatting	Strong sense of organization and finance, industry writing
CAD	Aircraft Modeling, Plan Manufacturing	CAD and materials knowledge, design proficiency
Manufacturing	Organize and Run Manufacturing	Laser cutting, composite lay-up, and 3D printing expertise
Systems	Payload Design and Mission-Specific Systems	Design proficiency, CAD experience, configuration process expertise



2.2 Schedule and Major Milestones

The team's schedule is shown in Figure X. The leadership teams and subteams both meet weekly to review scheduling, assess changes, and complete tasks. The entire team meets every other week to provide updates and work through design challenges as a larger group, as well as writing as a collective.

3. Conceptual Design

3.1 Problem Statement

The 2025-2026 AIAA DBF competition challenges teams to design, build, and demonstrate a

banner-towing bush plane capable of executing a diverse mission profile representative of real-world aircraft operations. This year's objectives involve a ground mission and three operational profiles: a test flight, a charter flight with passengers and cargo, and a banner towing flight requiring safe deployment and release of the banner.

3.1.1 Mission Scoring

Prior to completing missions, the team's aircraft must pass a thorough tech inspection to ensure compliance with all regulations and safe operation of the aircraft. Maximum payload and banner length are declared during tech inspection and must be observed in all applicable missions. During staging, the team will be given five minutes to install all batteries and payloads, verify control systems, and prepare the aircraft before entering the flight line. Overall team scoring has five basic components, which are Mission 1 (M1), Mission 2 (M2), Mission 3 (M3), Ground Mission (GM), and Report Score. These combine to give the total score as seen below:

$$Total\ Mission\ Score = GM + M1 + M2 + M3$$

$$Report\ Score = (0.85 \times Design\ Report\ Score) + (0.15 \times Proposal\ Score)$$

$$Overall\ Score = Report\ Score \times Total\ Mission\ Score$$

Equation (X)

3.1.2 Ground Mission – Passenger/Cargo Loading & Banner Demonstration

GM evaluates the team's ability to efficiently conduct the loading, unloading, and banner handling steps required for M2 and M3. GM is performed without propulsion batteries and consists of the following steps. First, the team will demonstrate functional flight controls. Next, the aircraft will be loaded with the declared passengers and cargo in their respective compartments with the proper restraints. Then, the team must unload the payload before installing the banner onto the aircraft. Finally, banner deployment and release will be demonstrated while the aircraft is held nose-up. After each step, the time is stopped so the ground crew can return to the start line. The ground mission score is based on time and is normalized as shown below:

$$GM = \frac{Time_{min}}{Time_{TAMU}}$$

Equation X

3.1.3 Mission 1 – Test Flight

M1 is designed to verify the basic airworthiness of the aircraft. The aircraft must complete three laps within a five-minute window. A successful landing on the paved runway is required. The scoring for this mission is pass/fail, as shown below:

$$M1 = 1 \text{ for successful mission}$$

Equation X

3.1.4 Mission 2 – Charter Flight

M2 assesses the aircraft's ability to generate revenue through carrying passengers (rubber ducks) and cargo (hockey pucks). The team must have at least 3 ducks for every 1 puck in the cargo, and is limited by the maximum declared capacity during tech inspection. The mission performance depends on the payload capacity, the number of laps completed within five minutes, and the total propulsion battery capacity. The scoring for this mission is calculated by three main equations that combine to make a total score, as seen below:

$$Income = (\#p)(I_{p1} + I_{p2} \times laps) + (\#c)(I_{c1} + I_{c2} \times laps)$$

$$EF = \frac{Total\ Propulsion\ Battery\ Capacity\ (Wh)}{100}$$

$$Cost = laps \times (C_e + \#p \times C_p + \#c \times C_c) \times EF$$

$$M2 = 1 + \frac{Income - Cost_{TAMU}}{Income - Cost_{max}}$$

Equation X

3.1.5 Mission 3 – Banner Flight

M3 evaluates the aircraft's banner towing performance. The externally stowed banner must demonstrate reliable deployment, towing, and release. The banner must be deployed after completion of the first upwind turn, and the banner must be released from the plane once it passes the start line following completion of the final lap. Mission performance is based on the number of laps completed, banner length, and the Rated Airplane Cost. The scoring equation is given below:

$$RAC = 0.5 * b + 0.75$$

$$M3 = 2 + \frac{(laps \times \frac{Banner\ length}{RAC})_{TAMU}}{(laps \times \frac{Banner\ length}{RAC})_{max}}$$

Equation X

3.2 System Design Requirements

The subsystem design requirements were derived from the mission parameters described in Section 3.1 and are shown in Table X. These criteria are categorized by the major systems they pertain to and will be used to ensure the aircraft is optimally designed.

Table #: Mission Sub Systems Requirements

Mission Subsystem Requirements

Category	Req. No.	Requirement
Configuration	1	Aircraft wingspan must be between 3 ft and 5ft, inclusive.
	2	Aircraft must be fixed-wing.
	3	Aircraft must have a single passenger compartment and at least one cargo compartment.
	4	Banner must be stowed externally for take-off.
	5	Banner cannot touch the ground during take-off and cannot interfere with the performance of the aircraft.
	6	Aircraft TOGW must be less than 55 lb.
Aerodynamics	7	Aircraft must maximize the lift generated to carry more payload.
	8	Aircraft must be capable of steady level flight in all mission configurations.
	9	Aircraft must optimize L/D.
Payload	10	Aircraft must be capable of carrying at least 3 ducks for every puck declared in tech inspection.
	11	Passengers must be securely restrained in an upright orientation.
	12	Passengers and cargo must be loaded into the aircraft quickly without the use of a speed loader.
	13	Passengers and cargo must be arranged in a way that minimizes the fuselage's size to reduce drag.
Banner	14	Banner must have a minimum length of 10 inches and a maximum aspect ratio of five.
	15	Banner must be lightweight and tear-resistant.
	16	Banner must remain vertical during flight once deployed.
	17	Banner design should minimize drag.
Propulsion	18	Aircraft must be propeller driven and electric powered.

	19	Total stored energy must not exceed 100Wh.
	20	Battery must be of a single type (LiPo or NiMh or NiCad).
	21	Batteries must be commercially produced Nickel-Cadmium, Nickel-Metal Hydride, or Lithium-Polymer batteries.
	22	Arming fuse must not exceed the maximum continuous discharge current of 100A.
	23	Each propulsion system must be connected to only one battery pack. A propulsion system consists of one motor, one external accessible arming fuse, one or more ESCs, and one battery.
Structures	24	Airframe must be lightweight yet strong enough to handle the payload weight at the maximum predicted aerodynamic loading.
	25	Aircraft's wings must be capable of withstanding the wingtip loading test.
	26	Aircraft doors and hatches must be securely fastened during flight.
	27	Banner deployment mechanism must be securely attached to the airframe.

3.3 Sensitivity Analysis

A sensitivity analysis was performed to quantify the effect of key scoring parameters on overall mission score and to guide design prioritization. The analysis was implemented using a Python scoring script that directly evaluates the M2 and M3 mission scoring equations outlined in Sections 3.1.1-3.1.5. M1 was excluded from this analysis due to its non-variable scoring structure.

For each mission, a baseline configuration was defined using nominal design values. Individual parameters were then independently varied over a defined range from the baseline while the other variables were held constant. The resulting change in mission score was recorded and normalized, allowing direct comparison of parameter sensitivity based on slope magnitude. Parameters with steeper slopes were identified as having a greater influence on score.

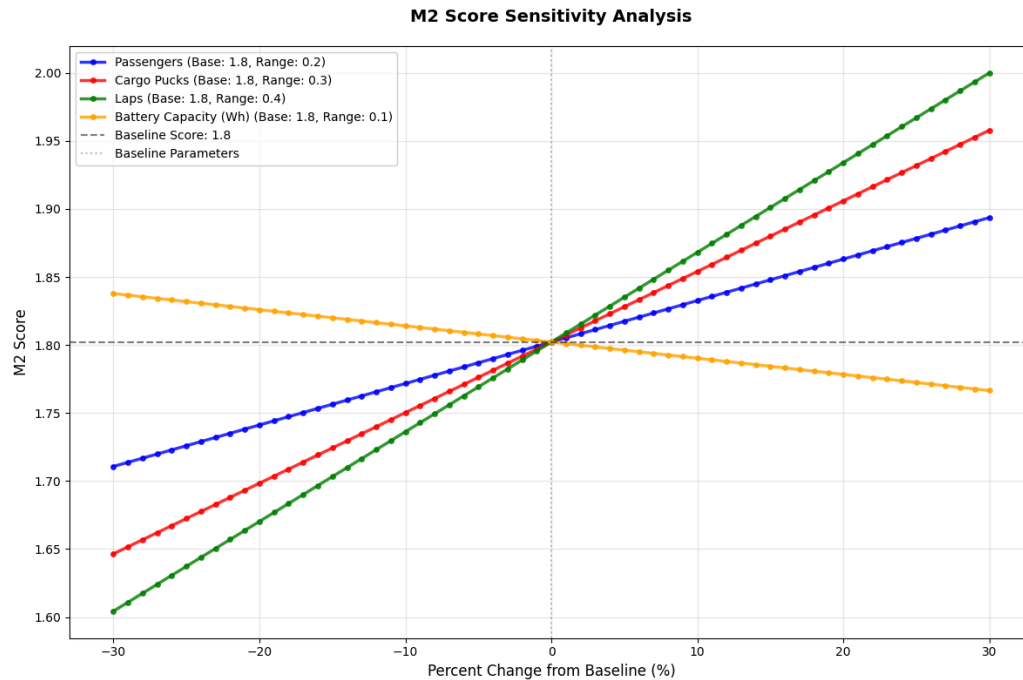


Fig X: M2 Score Sensitivity

Figure X displays the M2 sensitivity results, where payload mass (lb), number of laps, and number of passengers exhibit the largest positive influence on score. Battery capacity (Wh) shows a comparatively weaker effect due to diminishing returns at higher energy levels. These results indicate that maximizing payload related parameters in M2 provides the greatest scoring benefit.

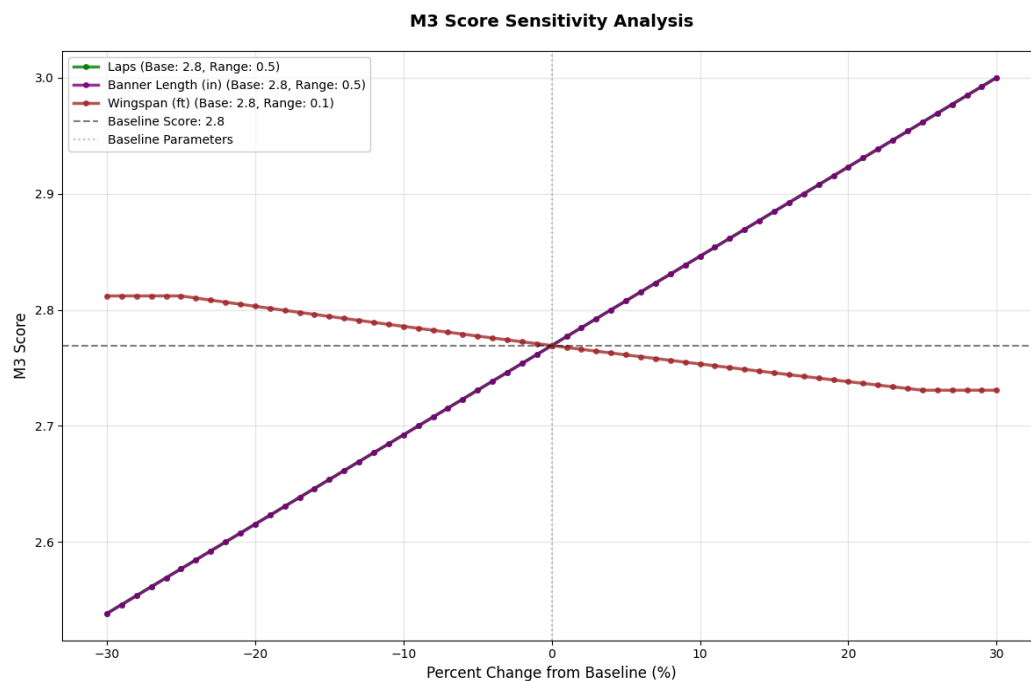


Figure X: M3 Score Sensitivity

Figure X shows the M3 sensitivity analysis. The dominant contributors to M3 score are number of laps and banner length (in). Wingspan (ft) exhibits comparatively weak sensitivity and is therefore treated as a design constraint rather than a primary optimization driver. As a result, the M3 design emphasis was placed on reliable lap completion and banner deployment rather than geometric optimization.

Based on these results, the design strategy prioritizes minimizing M2 lap time while maximizing M2 payload capacity, as this combination produces the largest increase in total score. For M3, emphasis is placed on achieving reliable lap completion and banner deployment. These conclusions are consistent with the trends observed in the sensitivity plots and with the relative weighting of parameters in the competition scoring equations. Using these conclusions, the team developed a constraint analysis algorithm detailed in Section 4.3.2. This was then incorporated into a coupled scoring sensitivity analysis program.

This program was created to better understand how design trade-offs would influence overall performance between M2 and GM. The M3 requirements are significantly less straining than M2, therefore, it was excluded from this coupled analysis. Instead of varying one parameter at a time, this analysis examines how changes in passenger count, puck count, and battery capacity interact to affect the combined mission score between M2 and GM while remaining within the constraints identified earlier. The results of this analysis are shown below in Figure X.

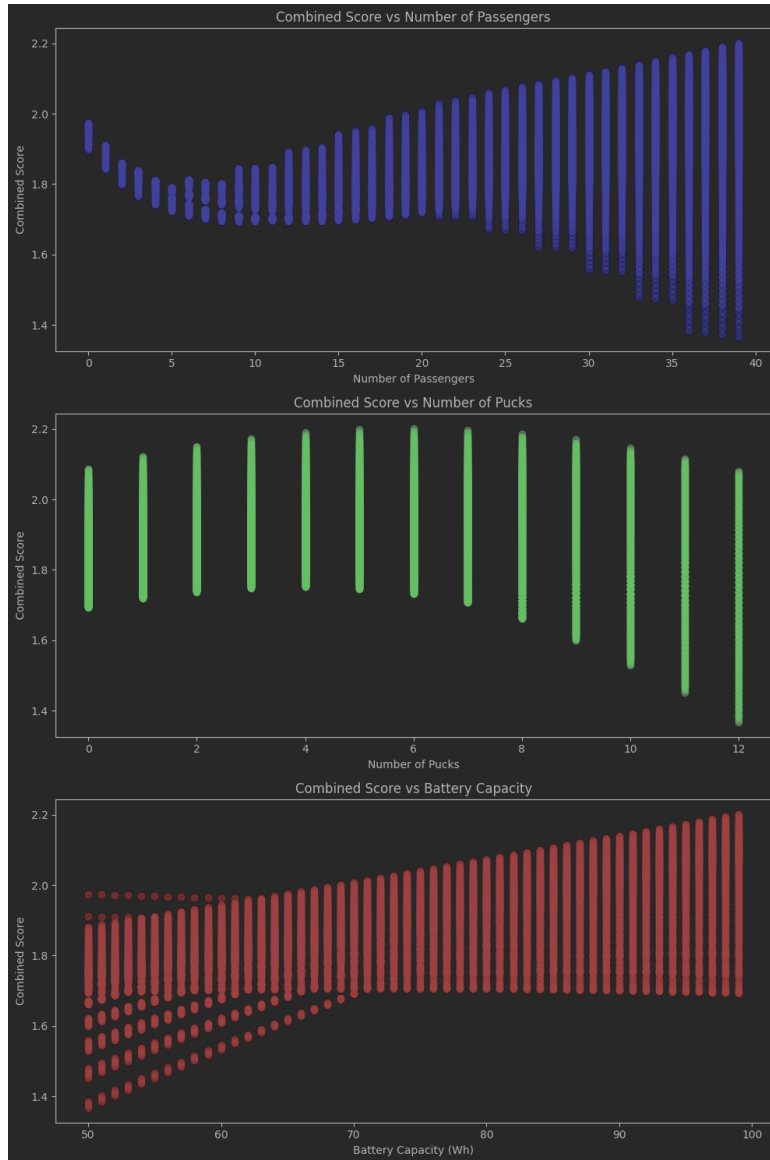


Figure X: Combined Score Sensitivity

Figure X presents the combined mission score of M2 as a function of passenger count, number of pucks, and battery capacity. The passenger sensitivity plot shows that increasing the number of passengers initially reduces the combined score due to added mass and drag. As passenger count increases further, the scoring benefit begins to offset these effects, leading to a gradual recovery in score. Based on this trend, a passenger count of 39 ducks was identified as optimal, providing strong scoring performance without excessive sensitivity to secondary design variables.

The puck sensitivity plot exhibits a peaked trend. Combined score increases as puck count rises from low values, reaches a maximum, and then decreases as additional pucks introduce mass and performance penalties that outweigh their scoring benefit. From this behavior, 6 pucks was identified as the optimal trade point, where scoring gain and mission performance are best balanced.

The final coupled analysis showed that the strongest overall performance occurs at 6 pucks and 39 ducks. Earlier versions of the model suggested higher puck counts, but this was driven by an optimistic battery efficiency estimate of 0.012 laps/Wh. After refining the energy consumption model to a more realistic value of 0.008 laps/Wh, the added energy penalty from carrying additional pucks became more significant, shifting the optimal point downward. As aerodynamic and structural constraints were refined through iteration, the script was updated to reflect these changes, ultimately converging on the 6 puck, 39 duck configuration. This configuration produced a maximum combined score of 2.079. For comparison, a GM focused configuration with a minimum number of pucks and ducks produced a maximum combined score of 1.837, reinforcing that the selected payload balance provides the most competitive overall result.

3.4 Trade Studies

The team conducted trade studies for each major aircraft system to select between competing configurations. For each system, a set of design criteria was defined and given importance based on a 1-5 scale, with 1 being least critical and 5 being most critical. Design options were then evaluated against these criteria, assigned performance scores, and combined into a total weighted score. The evaluation criteria and scoring scales are shown below.

Weight Scale

- 5: Very Important
- 4: Important
- 3: Moderately Important
- 2: Slightly Important
- 1: Not Important

Performance Scale

- 5: Greatly Exceeds Standards
- 4: Exceeds Standards
- 3: Meets Standards
- 2: Approaching Standards
- 1: Below Standard

3.4.1 Aircraft Configuration

Delta wing, blended wing body, conventional, and swept wing designs were considered for the aircraft's wing layout configuration. The analysis of these configurations is compared in Table (x). The delta wing was examined in preliminary design due to the large wing area to maximize lift for a relatively short wingspan and increased longitudinal static stability. However, due to the challenging manufacturability and poor lateral stability, this configuration was not chosen. The blended wing body configuration was found to provide the most wing area, maximizing the amount of lift the aircraft can produce, but had the same deficiencies as the delta wing in greater magnitudes and was hence not selected. For these reasons, the swept wing configuration was initially chosen due to the greater lateral stability.

The placement of the wings on the fuselage was largely constrained by the requirements of M2. Since the passengers and cargo have non-negligible masses, it was determined that their aggregate CM should be placed as near to the CM of the unloaded plane as possible. This means that they should be placed either over or under the wingbox (as it is infeasible to place them within). To minimize the cross-sectional area (and therefore drag) of the fuselage, it was determined that only high- and low-wing designs would be considered. Since the passengers and cargo must be in separate, nonremovable bays, the team decided that it would be best to load them via hatches. These hatches could either be on the sides, the bottom, or the top. As it is generally easier to load things from the top than it is from the bottom, the low-wing design was found to be the best option.

After the initial swept wing configuration was decided upon, MK1 was designed, built, and flown. During flight, the pilot and members observed issues relating to dynamic characteristics. These characteristics had been considered, utilizing XFLR5. The characteristics required re-evaluation of the configurations, and resulted in observations that delta wing, blended wing body, and swept wing configurations tended toward poorer dynamic characteristics than a conventional wing. After re-evaluation, the team decided to move forward with a conventional configuration.

Wing Configuration					
Criteria	Importance	Delta Wing	Blended Wing Body	Conventional	Swept Wing
Longitudinal Stability	3	5	5	4	4
Lateral Stability	2	2	1	4	5
Manufacturability	4	3	1	4	3
Wing Area	5	5	5	4	4
(Revealed) Dynamic Characteristics	4	2	1	4	3
Total		64	50	72	66

3.4.2 Fuselage Configuration

A total of four fuselage configurations were compared: a lifting body fuselage, a conventional, rectangular fuselage, a boom fuselage, and a double boom fuselage. Each configuration was graded on four criteria: added lift, usable volume, manufacturability, and mass. The weights, as well as each design's scores relative to the conventional configuration, were decided based on mission requirements.

The lifting body configuration excels at providing extra lift, but has limited usable volume, is difficult to manufacture, and is heavier than the conventional. The boom configuration provides less lift than the conventional and has much less usable volume. However, it is more manufacturable and has a smaller cross sectional area than the conventional. The double boom has worse manufacturability and is slightly larger, but is otherwise identical to the boom. When the configurations were compared by their weighted characteristics, it was determined that the conventional fuselage was the best choice for this set of missions.

Fuselage Configuration					
Criteria	Importance	Lifting Body	Conventional	Boom	Double Boom
Lift	2	5	4	3	3
Usable Volume	4	2	4	1	1
Manufacturability	4	1	4	4	2
Mass	2	3	4	5	4
Total		28	48	36	26

3.4.3 Tail Configuration

Conventional, mid-tail, t-tail, and v-tail designs were compared based on four weighted criteria: drag, stability, manufacturability, and mass. Each of the configurations' scores, as well as the weights of the criteria, were decided based on reference RC aircraft. The conventional configuration was taken as a standard. The mid-tail design has similar drag, but worse stability, is more difficult to manufacture, and adds more weight. The T-tail design is worse than the mid-tail in terms of stability and manufacturability, and, while smaller than the mid-tail, is still larger than the conventional. The V-tail configuration has less drag and comparable mass, but is less stable and less manufacturable. When each design was compared by their weighted characteristics, it was determined that the conventional tail was the best choice for the set of criteria.

Tail Configuration					
Criteria	Importance	Conventional	Cruciform	T-Tail	V-Tail
Drag	2	3	4	4	5
Stability	4	4	3	2	3
Manufacturability	5	4	2	2	3
Mass	1	2	3	3	4
Total		44	33	29	41

3.4.5 Landing Gear Configuration

Landing Gear Configuration					
Criteria	Importance	Tricycle	Tail Dragger	Tip Tail Dragger	Tandem with Outrigger
Drag	1	4	1	3	2
Manufacturability	2	2	4	3	1
Tail Strike Risk	3	1	3	3	2
Weight	4	3	1	2	3
Ground Stability	5	4	1	2	3
Total		43	27	36	37

Based on the decision matrix in table X, the team chose a tricycle landing gear configuration. A trade study was conducted on four common landing gear designs to guide this decision. Of the criteria, ground stability is the most important criteria as it allows the aircraft to take off and land safely, which is essential for a successful flight mission. Lowering the weight allows the plane to carry more cargo, making it the second most critical factor. Drag and manufacturability are weighted least as their overall impact is negligible. The tricycle landing gear offers the significant benefits of ground stability and its low weight that nullifies the disadvantages of a higher tail strike risk and manufacturing complexity.

3.4.6 Propulsion Configuration

Propulsion Configuration			
Criteria	Importance	Single Tractor	Twin Tractor
Cost	1	5	4
Manufacturability	2	5	4
Stability	3	4	5
Weight	4	5	4
Thrust Efficiency	5	5	4
Total		72	63

The trade study indicates that a single tractor plane is the optimal design. The propulsion configurations that were evaluated were single tractor and the twin tractor, as all pusher-props were deemed incompatible with the banner release. The criteria were chosen from analysis of mission objectives. Thrust efficiency is the most important factor because it facilitates higher scoring by increasing flight time. The lower weight of a single motor and its high Thrust/Weight ratio maximizes laps flown by enabling a higher average airspeed. A single tractor also offers advantages over a twin tractor in terms of manufacturability and cost. These factors validate the single tractor as the preferred configuration.

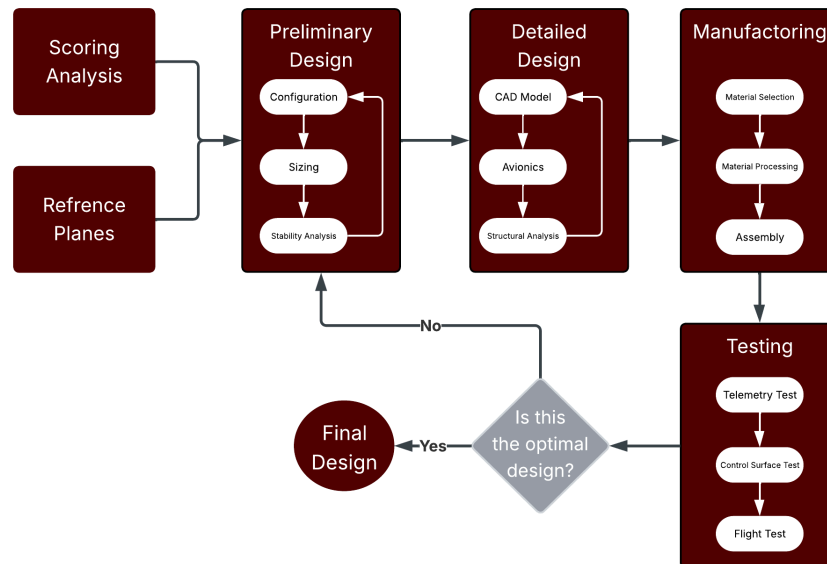
4 Preliminary Design

The preliminary design of the aircraft was an iterative process of sizing equations, vortex-lattice methods, and computational fluid dynamics simulations to arrive at an optimal design for stability and performance.

4.1 Design Methodology

Texas A&M's design methodology was built on techniques from past years and a more in-depth understanding of CFD. The score sensitivity analysis provided in section 3.3 shows that the aircraft should be optimized for M2 to have the highest scoring potential. With the determined optimal value of 6 pucks and 39 ducks, along with the weights from previous TAMU DBF aircraft, an initial maximum takeoff weight estimate of the aircraft was determined to be 20 pounds. Aircraft velocity for M2 and M3 was then calculated using estimates of the plane's battery capacity. These values were used to perform a constraint analysis, determining the best optimized design points for the aircraft. Following constraint analysis, the aerodynamic design of the aircraft was initiated with airfoil analysis. Once the airfoil was selected, the optimal wing planform was selected through an iterative sizing process within XFLR5. Tail sizing was completed to achieve static stability targets, followed by the use of aircraft inertia estimates to verify dynamic stability in accordance with MIL-F-8785C standards. A CAD of the aircraft's shell was produced based on the initial sizing. The shell CAD was then put through StarCCM+ to confirm lift, drag, and stability values. Any needed modifications were validated in XFLR5 before being confirmed in CFD again. Once all necessary modifications were made, and checked against all requirements stated in section 3.2,

internal structures and avionics components were modeled using CAD. Structural analysis was performed on these designs, and optimized to increase the strength and weight of the aircraft structure. Manufacturing began once the detailed design was completed. This iterative process is illustrated in Fig. # below.



4.2 Mission Model

To predict the performance of the proposed aircraft and to verify compliance with mission requirements, a physics based mission model was developed in Python. The model simulates the full flight profile of a mission lap by numerically integrating the equations of motion using aerodynamic, propulsion, and mass properties of the aircraft. Aerodynamic lift and drag coefficients, obtained from precomputed airfoil and configuration data, were implemented through interpolation with respect to angle of attack. Propulsion performance was represented using a thrust model that accounts for throttle setting and thrust decay with increasing airspeed.

The mission simulation is divided into distinct flight phases: take off and climb out, straight and level flight segments, and coordinated turning maneuvers. Each flight phase was modeled using either longitudinal dynamics, or planar kinematics. This approach reduces computational complexity while retaining the dominant performance drivers relevant to the preliminary design.

During the take off phase, rolling friction and aerodynamic drag is taken into account when simulating the acceleration of the aircraft on the ground at full throttle. Rotation is initiated at a prescribed rotation speed, after which the aircraft pitches to a fixed angle of attack until sufficient lift is generated for lift off. The aircraft then climbs at a constant climb angle until a target altitude of 200 ft is reached, marking the end of the climb out segment.

Straight flight segments are modeled at a constant target airspeed using reduced throttle and a fixed angle of attack representative of near level flight. Turning segments are simulated using a prescribed turn

radius and heading rate, with the resulting change in heading applied incrementally to capture the aircraft's curved flight path. Although each segment is evaluated in two dimensions, the full mission trajectory is reconstructed by sequentially combining the individual segments, resulting in a spatial flight path that captures the overall mission geometry.

The resulting mission flight path is shown in figure X, which illustrates the sequence of straight segments, turns, and altitude changes throughout a mission lap. This figure is generated directly from the mission model and is used to visually confirm mission layout, turn geometry, and altitude compliance. The mission phases depicted in figure X correspond directly to the segments described in this section and form the basis for subsequent performance evaluation.

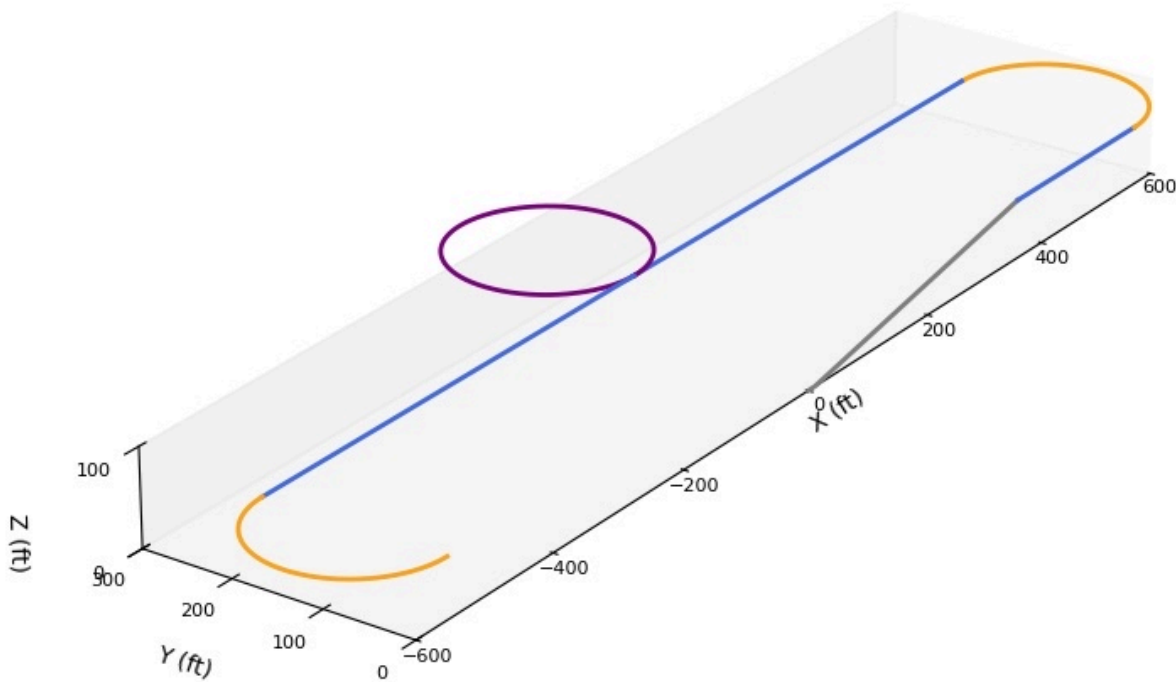


Figure X: Simulated Mission Lap

While the mission model captures the primary performance trends, several simplifying assumptions were applied. These consisted of sea level standard density, negligible control system dynamics, and an idealized thrust curve of the propulsion system. Despite these limitations, the model provides sufficient fidelity for preliminary design trade studies and enables consistent comparison between configuration changes.

4.3 Aerodynamic Design Analysis

The aerodynamic design of the TAMU DBF aircraft started with airfoil analysis, then moved into wing design. After the wing was complete, the aircraft's tail was sized to optimize static stability. Inertia

estimates from CAD then allowed the team to confirm the aircraft's dynamic stability, before the design's static stability was fully confirmed using high fidelity CFD simulations.

4.3.1 Weight Estimation

The weight of the aircraft was estimated based on payload weight as well as previous TAMU DBF aircraft. Using the coupled score sensitivity analysis results, estimates of avionics weight, and structures weight from previous aircraft, the estimated weight breakdown is presented in table XX.

Table XX. Initial Aircraft Component Weight Estimates

Initial Aircraft Component Weight Estimates	
Component	Weight (lb)
Structural Components	12
Motor and Propeller	0.97
Battery	1.275
Avionics	0.60625
Maximum Banner Weight	0.1875
Puck Weight	2.25
Duck Weight	1.5844
M1 Weight	14.8513
M2 Weight	18.6856
M3 Weight	15.0388

The structural components weight estimates (wings, fuselage, tail) were based on density estimates of previous TAMU DBF aircraft. Based on the values presented in the table above, a factor of safety was applied, and a maximum weight of 20 lbs was chosen. The final weights of the aircraft would be determined after the preliminary design.

4.3.2 Constraint Analysis

In order to find an initial design point for the aircraft, the team determined that a constraint analysis would be done in order to find the optimal T/W and W/S ratios. By analyzing the mission requirements, the following constraint equations were applied to the prospective aircraft.

T/W for a Desired T-O Ground Run Distance

$$\frac{T}{W} = \frac{1.21}{g\rho C_{L_{max}} \cdot S_G} \left(\frac{W}{S} \right) + \frac{0.605}{C_{L_{max}}} (C_{D_{TO}} - \mu C_{L_{TO}}) + \mu$$

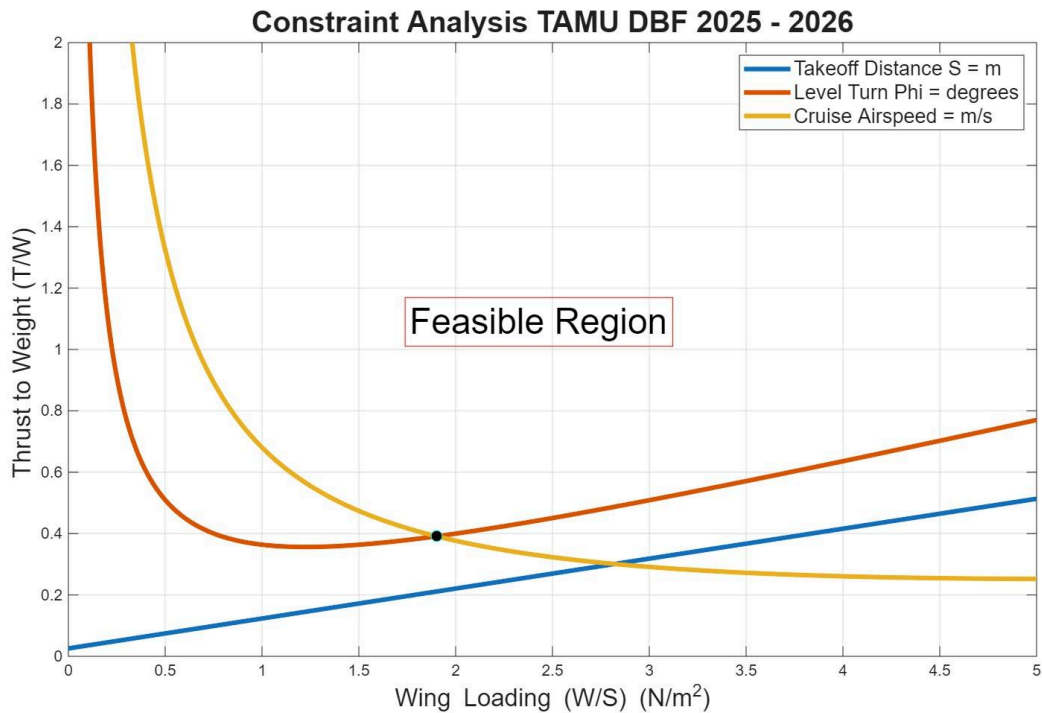
T/W for a Desired Cruise Airspeed

$$\frac{T}{W} = q C_{D_{min}} \left(\frac{1}{W/S} \right) + k \left(\frac{1}{q} \right) \left(\frac{W}{S} \right)$$

T/W for a Level Constant Velocity Turn

$$\frac{T}{W} = q \left[\frac{C_{D_{min}}}{(W/S)} + k \left(\frac{n}{q} \right)^2 \left(\frac{W}{S} \right) \right]$$

Each of the equations were chosen as strictly necessary for the performance of the aircraft. The projected flight path has only straight climbs and level altitude turns with constant velocity. The constraint analysis was performed with the requirements in line with the selection of the powertrain having been picked and thus giving a target airspeed for each of M2 and M3. The analysis was performed for both M2 and M3, with the constraints from M3 being more constrictive than those of M2, due to the added weight.



The graph in **Figure X** shows that the feasible region has an optimized point located at a thrust-to-weight ratio (T/W) of 0.391 and a wing loading (W/S) of 1.902 lb/ft². To account for modeling uncertainty and to ensure compliance with safety requirements, a safety factor of 1.3 was applied to the baseline thrust-to-weight ratio, resulting in a final design value of T/W = 0.509. The updated design point represents a feasible and compliant performance region. These values were then used as the foundation

for the coupled scoring sensitivity analysis between GM and M2, ensuring that payload optimization was performed around a realistic and physically achievable aircraft configuration.

4.3.3 Airfoil Selection

The aerodynamic analysis of the aircraft began with the airfoil selection process. The team decided to use XFLR5 to analyze the airfoils at various low Reynolds numbers. The primary objective of this analysis was to assess the lift and drag characteristics of different airfoils and their relationship with angle of attack. It was necessary for the airfoil selected to produce 20 pounds of lift in trim in order to meet the maximum takeoff weight requirement. It was also important that the airfoil had a negative coefficient of moment to ensure the aircraft's static stability. The initial analysis used XFOIL within the program with airfoil splines downloaded from Airfoil Tools to perform a 2D analysis. This provided a rough batch of airfoils to further analyze in 3D using Lifting Line Theory. From the data gathered from simulations, the selection was narrowed down to 5 airfoils: NACA 4412, Clark Y, Eppler 214, Eppler 193, and N22. A decision matrix was created to select the airfoil, and is presented in table OO.

Table OO Airfoil Decision Matrix

Wing Airfoil Decision Matrix						
Criteria	Importance	N22	NACA 4412	Clark Y	Eppler 214	Eppler 193
Manufacturability	1	4	5	2	1	1
Stability	2	4	4	5	1	1
Efficiency (C_l/C_d)	3	2	3	2	4	4
Lift at 0 angle of attack	4	5	3	4	2	2
Total		38	34	34	23	23

4.3.4 Wing Design

Given the results from score sensitivity analysis showing that wingspan was a less important factor in score, along with the fact that a higher aspect ratio meant a more efficient aircraft, the team decided to keep a 5 foot wing span. Various wing tapers, sweeps, and twists were analyzed in XFLR5 to find the most optimal wing planform. Leading edge sweeps from 0° – 60° , wing taper ratios from 1 – 0.2, and wing twists from 0° – 3.5° were swept through. The MK1 design had found that the optimal design that would maximize L/D would have 50° of leading edge sweep. However, it was found that the high sweep increased the aircraft's C_{lB} value too much, causing a high amount of dutch roll. This design also caused various structural issues, resulting in the design being abandoned in favor of a more stable and easier to manufacture design for MK2. To maximize lift at the lower speeds that would be required to tow the banner, the team opted for an unswept wing. While this design increased drag compared to MK1, the increased C_l value allowed for the plane to produce more lift, increasing the banner size that the plane could fly with. With the straight wing planform selected, the team next analyzed wing twists and washout to improve tip stall characteristics and increase the aircraft's lift. It was found that a twist of 1.5° at the root was necessary in order for the wing to have enough lift at the slower velocities planned for the M3 banner

flight. The team also experimented with various wash out angles to improve tip stall characteristics, and it was found that reducing the twist angle to 0° at the wing tips provided improved stall performance while maintaining enough lift at slower speeds. Various taper ratios were also tested, however it was found that an increased chord length was required in order to maintain lift, which caused increased drag. While adding wing taper did improve the aircraft's low C_{La} value caused by the low aspect ratio, the team decided that the drag improvements were not worth the loss in lift caused by the taper. The taper also reduced the wing area, which had the potential to cause stability issues during takeoff and landing. For these reasons, the team opted for a straight untapered wing with washout. Additional analysis of the aircraft's static stability found that the aircraft's C_{IB} value was significantly below a stable value, thus dihedral was added to the wing to increase roll stability. A summary of the wing sizing has been provided in table XX.

Wing Size Summary

Wing Size Summary	
Parameter	Value
Sweep	0°
Taper Ratio	1
Span	59 inches
Chord	17 inches
Root Incidence	1.5°
Tip Incidence	0°
Aspect Ratio	3.471
Dihedral	3°

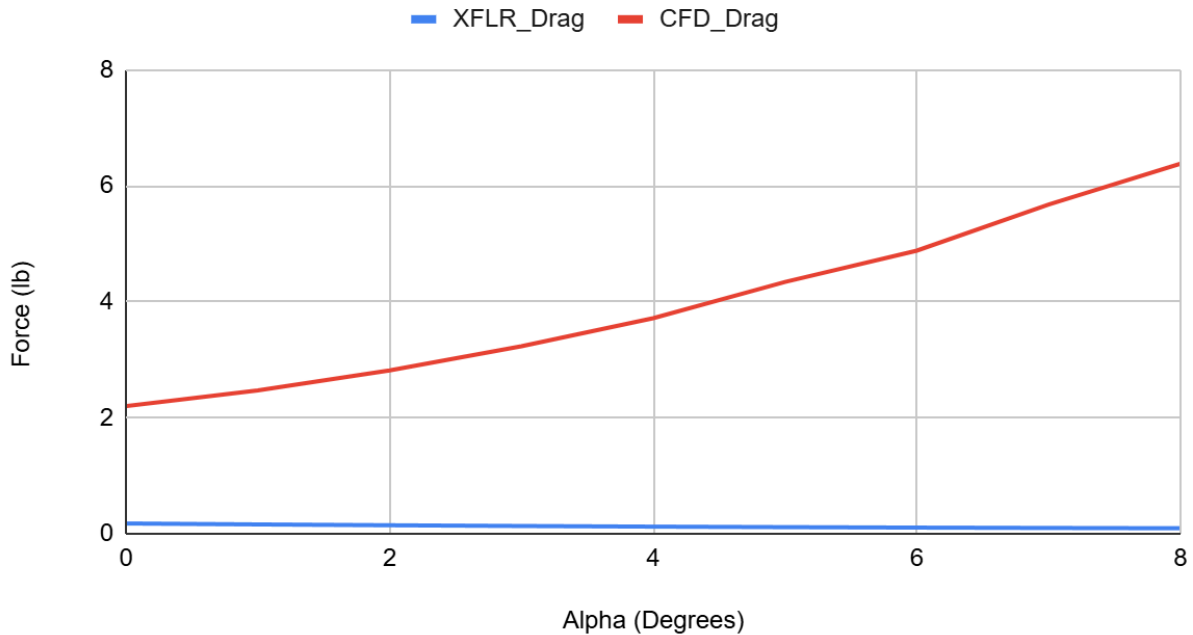
4.3.5 Drag Analysis

The primary purpose of performing a drag analysis was to design for the optimal M3 performance, as it was the mission that involved the highest amount of drag forces on the aircraft. This started with the airfoil selection. The N22 was chosen because it provided the highest L/D ratio at low speeds, which allows the plane to keep flying while towing a larger banner. The banner itself underwent rigorous drag testing to understand the overall forces contributed by it. This allowed the team to attempt to maximize banner size using formulaic methods. The tests resulted in the data shown in Section 4.3.8.

However, prior to fixing a banner, the team needed to know the value of the drag from the aircraft itself. This analysis was done using both XFLR5 and STAR-CCM+. The XFLR drag analysis was only able to indicate the wing performance. It returned a value of 0.19 lbs at $\alpha = 0$ for the Mk 2 design. However, the majority of the drag would be due to the fuselage, which is where STAR-CCM+ was used. The team performed an alpha sweep of the aircraft to derive coefficients and numbers for the drag on the aircraft.

Figure X shows the drag curve for positive AoAs up to 8 degrees.

XFLR vs. CFD Drag



According to this information, the drag force at $\alpha = 0$ is 2.2 lbs, with a drag slope of approximately 0.526 lb/degree. The optimization for the banner, as previously stated, is located in Section 4.3.8, in which this data was used.

4.3.6 Tail Sizing

Once the wing planform had been selected, the tail geometry was sized and analyzed in XFLR5. Based on previous TAMU DBF aircraft, an optimal static margin of 8% was chosen since it provided high maneuverability while still being controllable by the pilot. Based on this estimate, along with XFLR5's estimate for C_{L_α} , the team was able to determine an optimal value for C_{m_α} that would provide optimal longitudinal stability. This was determined using Eq. XX.

$$C_{m_\alpha} = C_{L_\alpha} \cdot (x_{CG} - x_{AC})$$

Since this aircraft needs to carry a large payload with a limited wingspan, the selected wing has a low aspect ratio, causing a reduced C_{L_α} value. This reduced value in turn reduced the needed C_{m_α} value.

Based on previous TAMU DBF aircraft design cycles, it is known that the fuselage has a significant destabilizing effect on the aircraft's overall stability. Since XFLR5 generally provides inaccurate and inconsistent values when fuselages are included, the tail sizing in XFLR5 was completed with a reasonable factor of safety to ensure the aircraft would not turn out to be too unstable when analyzed with fuselage effects in CFD. An initial C_{m_α} target value was chosen to be -0.5 rad^{-1} based on previous TAMU DBF aircraft. Eq. XX, the longitudinal buildup equation from [Roskam, Volume 3] was then used to find an optimal tail size for the aircraft.

$$C_{m_{cg\alpha}} = C_{LW_\alpha}(X_{CG} - X_{AC}) - \eta\left(\frac{S_H}{S_W}\right)C_{LH_\alpha}\left(1 - \frac{d\varepsilon}{d\alpha}\right)(X_H - X_{CG})$$

By assuming a tail efficiency factor (η) of 1, and neglecting downwash effects, the buildup equation allowed for the determination of the horizontal stabilizer area. The initial size provided by this equation was then iterated on in XFLR5 to have a reasonable fuselage length while maintaining both static and dynamic stability. Lateral stability was completed using a similar process, with a $C_{n\beta}$ target set at 0.1 rad^{-1} based on previous TAMU DBF aircraft. The vertical stabilizer was analyzed with various taper ratios to find the optimal value to be most manufacturable and to have the least drag, while still maintaining lateral stability. A NACA 0012 airfoil was selected for both tail surfaces to provide minimal drag while allowing easy manufacturability. After this initial sizing was complete, the dynamic stability analysis revealed that some modifications to the tail were necessary. The dynamic stability initially showed a level 2 flying quality for dutch roll, and thus, the vertical stabilizer span was increased. Additionally, phugoid mode was also significantly underdamped. Through iteration in XFLR5, it was noted that a lower horizontal stabilizer aspect ratio improved phugoid characteristics. For this reason, the horizontal stabilizer aspect ratio was reduced. The aircraft was then analyzed in CFD, and it showed that further modifications were necessary. The result of the CFD analysis showed that the static margin with the fuselage effects included was 8.43, very close to the target of 8%, however, the trim angle in CFD was significantly higher than predicted, at 3° instead of 0. For this reason, the horizontal tail incidence was increased to -2° . The lateral and directional stability from CFD closely matched the results from XFLR5, meaning that the vertical stabilizer did not need any further modification. The stability information has been expanded on further in Section 4.7, and the tail sizing is summarized in Table XX.

Tail Sizing Summary

Tail Sizing Summary		
Value	Horizontal Stabilizer	Vertical Stabilizer
Span	26 in	10 in
Root Chord	12 in	10 in
Tip Chord	12 in	5 in
Taper Ratio	1	0.5
Aspect Ratio	5.33	2.17
Airfoil	NACA 0012	NACA 0012

4.3.7 Control Surface Sizing

Control surfaces were sized to ensure the aircraft maintains adequate authority in pitch, roll, and yaw during the low speed portions of the mission, especially takeoff, climb, and turning flight. Surface geometry was adjusted iteratively and evaluated using aerodynamic control derivatives. For each surface, the change in aerodynamic coefficient was calculated with and without a small control deflection, and this difference was divided by the deflection angle to obtain the control derivative. The derivatives were evaluated at a zero angle of attack to represent the trimmed flight condition. Surface sizes were then adjusted until the aircraft demonstrated sufficient control response without requiring

excessively large surfaces. The obtained derivatives were then compared to similar aircraft, including the Ryan Navion as well as previous DBF Aircraft.

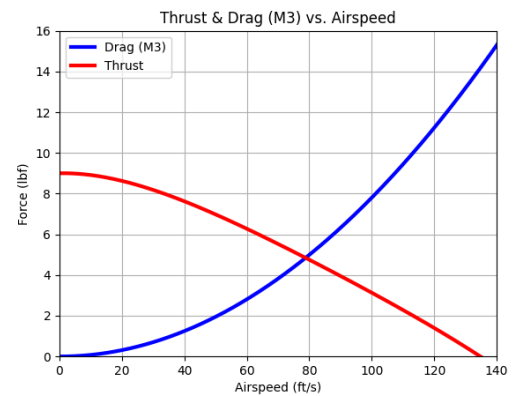
The resulting control derivatives are shown below in Table XX. The elevator provides adequate pitch authority for rotation and climb out, the ailerons provide a responsive roll rate for mission turns, and the rudder supplies enough yaw control to maintain coordinated flight. Overall, the selected control surface sizing provides reliable controllability while avoiding unnecessary drag or structural complexity.

Control Surface Sizing		
Control Surface	Stability Derivative	Value
Aileron	$C_{l_{\delta a}}$	0.211
Elevator	$C_{m_{\delta e}}$	-1.262
Rudder	$C_{n_{\delta r}}$	-0.080

4.3.8 Banner Design

To determine banner sizing and application, heavy influence was taken from the 2020 Design Build Fly competition, where teams similarly needed to deploy and release a banner. For initial sizing purposes, data from USC's banner testing was used to create a model to approximate banner size in M3. A significant finding from USC's data was that a banner aspect ratio of 5 provided the most efficient banner in terms of drag. The data found by USC was used to create an initial model in MATLAB for banner size as a function of velocity. The assumptions from USC's banner test were eventually replaced by data collected during TAMU DBF's own banner test, which is discussed in more detail in section XX. The collected data was used to determine a length for the banner.

First, mission 3 lift requirements were analyzed, and it was found that the aircraft could tow a competitive banner size at a reduced cruise speed of 75 ft/s. To determine the total drag force for M3, accounting for both the aircraft and the banner, equation X was derived as a function of banner length and airspeed. The estimated drag coefficients of the aircraft and the banner were collected from CFD and physical banner testing. From the data of the selected propulsion system, which will be discussed later in section 7.3.1, the team was able to generate the thrust curve while varying the length of the banner to find its maximum value that will achieve the required cruise speed. Fig. XX shows that when the plane is towing a 75 inch banner, it can cruise around 80 ft/s, which was chosen over 75 ft/s as an added factor of safety.



$$D_{total} = \frac{1}{2} \rho V^2 (c_{d_{aircraft}} S_{wing} + c_{d_{banner}} \frac{L_{banner}^2}{5}) \quad \text{Equation X}$$

Fig. XX Mission 3 Drag at a reduced speed

4.4 Mission Payloads

The aircraft design was optimized to meet the mission payload requirements for M2 and M3. Based on the aircraft performance and payload analysis data shown in figure **(Coupled Sensitivity)**, the aircraft is capable of carrying 5-7 hockey pucks (6oz/each) and up to 39 rubber ducks (0.7oz/each). This maximizes mission point potential while maintaining safe flight margins, with the large wing area allowing the aircraft to safely support high payload weights. The fuselage includes two large top-side hatches to allow access for both M2 payload compartments and M3 mechanism installation. The fuselage also includes internal 3D printed latches to connect to the payload compartments with half-height bulkheads providing support to the payloads. The fuselage bottom also includes a non-trussed section to provide structural strength for impulses caused by banner release, and the plane itself is capable of carrying and deploying a banner 70 inches in length without compromising flight performance.

4.4.1 Payload Configuration

The payload section will store the passengers and cargo in their respective compartments as shown below in figure **X**. The ducks will be secured using one board on both the bottom and the top of the compartment, with each board serving to constrain a different direction of movement. The bottom board, which is made of poplar, encloses the lower body of the ducks to prevent lateral movement. The top board, made of foam, conforms against the shape of the ducks' upper bodies to prevent vertical movement. The pucks will be stacked on top of each other and secured with a velcro attachment. The payload containers will be removable from the fuselage to improve performance in M1 and M3.

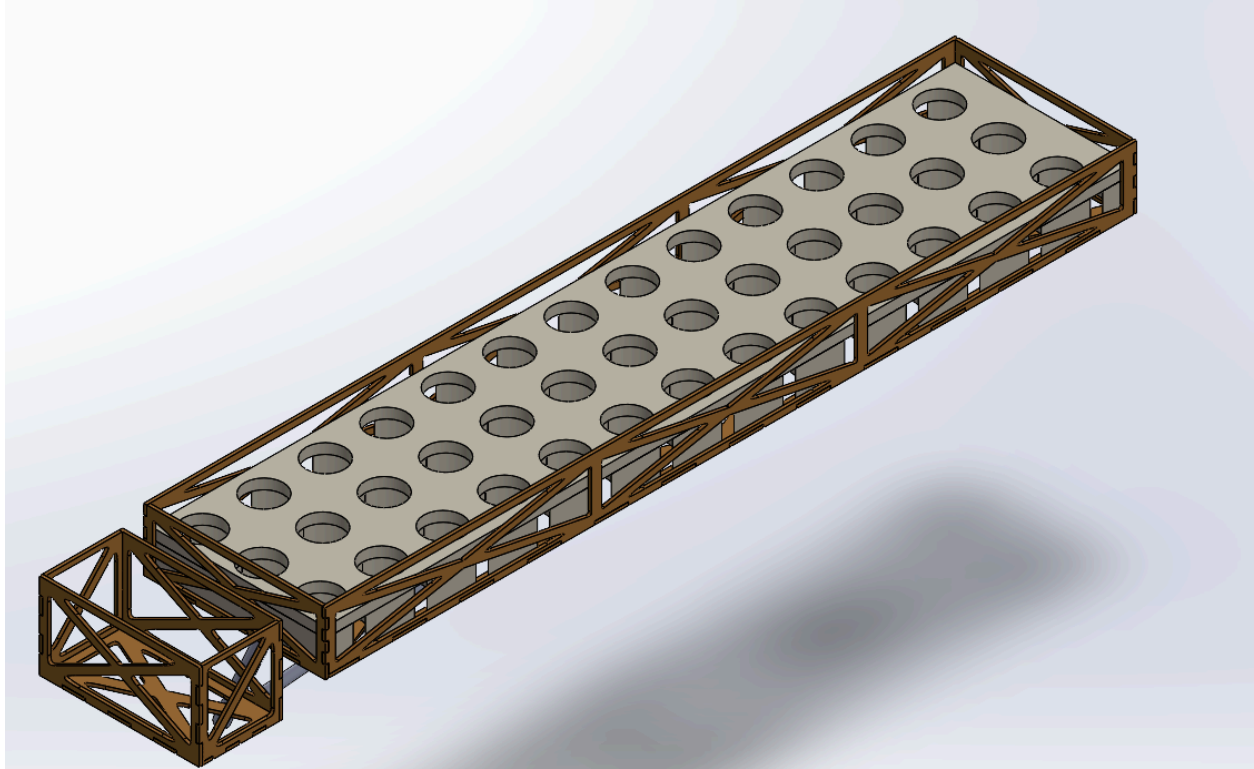


Table #. Decision Matrix for the payload configuration of the passengers.

Decision Matrix - Payload Configuration				
Criteria	Importance	Cover	Slider	Velcro
Loading Speed	5	5	2	3
Space Efficiency	2	5	5	4
Strength	3	4	3	2
Weight	4	3	4	3
Total		59	45	41

Table X shows various storage methods for the passengers within the aircraft. The considerations are loading speed, space efficiency, strength, and weight. The compartments must be quick to load and unload, as this will be done under a time constraint. The space efficiency of the design is also relevant, as packing the cargo efficiently can allow more passengers to be stored with less volume. However, it is less of a concern as the aircraft has ample room for the payload. The strength of the configuration is

moderately important, as the aircraft will experience high loads during operation. Weight is a crucial factor, as it increases the wing loading and the minimum velocity required for landing and takeoff. Loading speed is the most important, as it has a large and direct impact on GM scoring.

The “slider” option consists of a panel similar to the cover, but it slides in from the side instead of coming over the top. This option is not as strong as the “cover” due to the mandatory gaps for the sliding mechanism to function, but it is also fairly light for that same reason. The next option is “velcro”, which involves velcro straps that go over the passengers. This option would be the least weight efficient, and was found to be slow and unstable. The “cover” option involves a foam cover with circular holes that conforms around the necks of the ducks. This cover is easily loaded, weight efficient, and secured reliably, but difficult to manufacture and was thus chosen.

4.4.2 Banner Release

The banner release system stores the banner underneath the fuselage, deploys the banner upon command, and releases the banner upon command. The banner release mechanism is a two-step servo system that unfurls and releases the banner shown in figure X. As seen in figure X the banner release mechanism is on a 3D printed plate that is pushed into the fuselage, during the staging for M3. Prior to attaching the banner to the banner release mechanism, the banner is laid out flat with one end of the tether tied to the leading edge of the banner. To attach the banner to the release mechanism, the banner is rolled tightly, and the tether is wrapped tightly around its center without tying. This ensures that when the banner is released, it will unfurl vertically, preventing any unwanted interference. The end of the tether not connected to the banner is then looped through the second increment of the servo arm. Next, the shock cord is pulled across the banner, hugging the banner tightly to the bottom of the fuselage. Then the shock cord is looped through the first increment of the servo arm. When the servo pulls the pin attached to the shock cord to half retraction, the shock cord is released, allowing the banner to deploy and unfurl to its full size. When the servo’s pin is fully retracted, this tether is released, releasing the banner from the plane.



Figure X

Table X: Decision Matrix for the banner release mechanism

Banner Release Mechanism				
Criteria	Importance	Servo pin	Wheel pin	Two point
Strength	4	4	3	5
Simplicity	2	4	3	4
Manufactureability	5	5	4	3
Total		49	44	43

Table X shows various systems that could be used to deploy and drop the banner during flight. When choosing a banner release mechanism design, the strength, simplicity, and manufacturability of the design was considered. The banner release mechanism will be securing the banner to the plane while deployed. The drag from the banner will create a strong reaction force on the mechanism. It is important for the mechanism not to break during flight. The servo pin was chosen as the banner release system for its manufacturability and strength. The servo pin system involves a servo, 3D printed banner mechanism, and a servo arm. Since the banner mechanism is 3D printed, it is easy to redesign and manufacture quickly. 3D printing allows for the choice to use a high strength material such as PETG-CF. The main concern with the servo pin mechanism is that the servo arm may break. The servo arm used is a thin

metal rod, when the rod is moved by the servo, the rod will be pushed at an angle. This angle adds stress onto the rod in addition to the stress caused by the shock cord and banner tether.

The wheel pin design involves a wheel with a cylindrical extrusion connected to a servo, a 3D printed rod, and 3 holding bands. When the servo is rotated, the wheel will rotate the rod, releasing the holding bands incrementally. As the holding bands are released the banner will deploy, then drop on command. This mechanism is manufacturable as the wheel and the rod are both 3D printed. The only concern is that the rod would be too long to 3D print in the printers available. The second concern was for the strength of the system. The rod designed would be ABS polymer, however it would also need to be relatively thin. The force of the banner would potentially cause it to break under fatigue stress.

The two point design involved two servos and two points of contact of the banner with the aircraft. One servo would be connected to the string hugging the banner to the base of the aircraft. The other servo would be connected to the banner tether. This design was not chosen as it would be difficult to manufacture. Two servos, requires two plates to be inserted into the plane during the staging period. In addition, having two servos would lead to using another controller during flight. This would make piloting the aircraft more difficult.

4.5 Landing Gear

The design of the landing gear assumes an incidence angle of 3 degrees and a rear landing gear mount 15 degrees behind the CG, in alignment with Roskam's textbook. A schematic was developed to allow for calculation of the height and placement of the landing gear in figure X. These calculations were critical in validating the design and preventing risk of tailstrike. The critical angle of attack was determined to be 13 degrees, so a tail strike angle of 12.77 degrees was set. By having the tail strike angle be slightly lower, the plane is able to tail strike before it stalls, allowing the pilot to maintain control of the plane in worst case scenarios. Struts were added as a weight efficient method for ensuring the structural integrity of the landing gear. The collar allows the front landing gear to rotate and still be securely fastened by the struts, as shown in figure Y. The front landing gear height was 13in to provide propeller clearance, with some margin for mounting purposes.

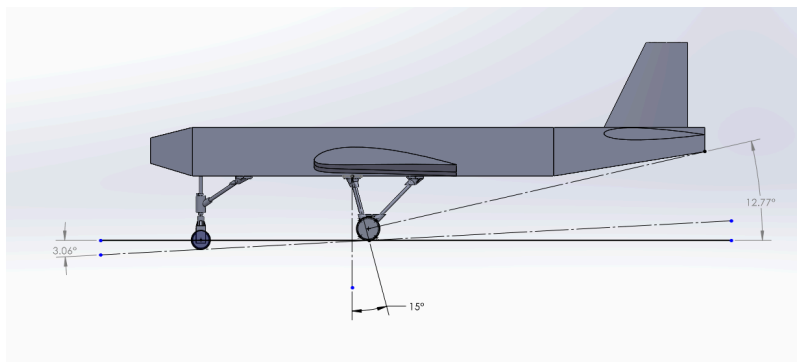


Figure Y

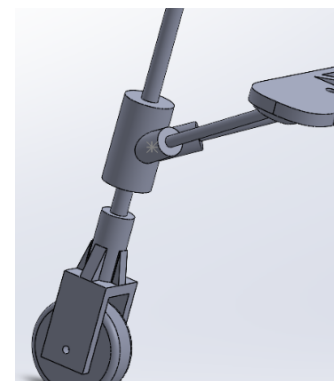


Figure X

The rear landing gear required the consideration of wheel position relative to the CG, landing gear width, angle of the wheels, and rod lengths. The rear wheel was placed such that the point of contact was 15 deg behind the CG. The rear landing gear width was chosen for the wheels to stick out past the fuselage slightly to help with ground stability and rolling when landing, without becoming a hindrance to landing and take off performance. The angle of the wheels was changed from a neutral position to a positive camber and a positive canted position at 3 degrees each in order to increase the stability and control in landing, as shown in figure Y. These decisions come with the downside that the wheels can't support as much weight as a neutral position wheel, additionally turning/taxing is harder but these were decided to be cost effective decisions.

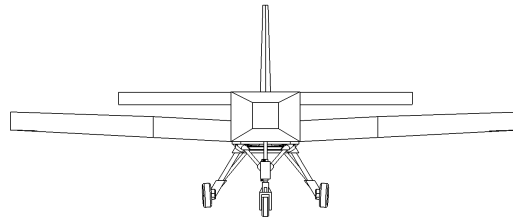


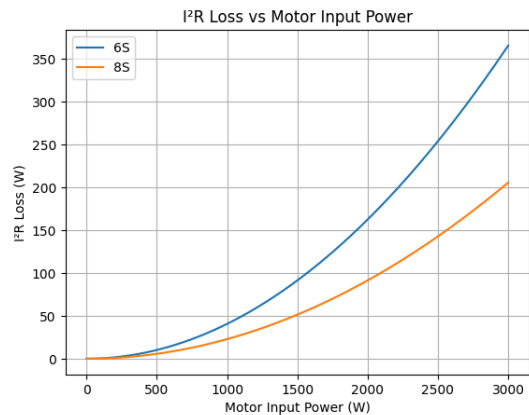
Figure X

4.6 Propulsion Analysis

4.6.1 Powertrain Selection

The rules specify that batteries must be commercially produced Nickel-Cadmium, Nickel-Metal Hydride, or LiPo. LiPo batteries were chosen over other batteries due to the significant weight reductions afforded by their higher energy density. The scoring sensitivity in Section 3.3 determined that increasing the number of laps flown improves scoring more than reducing battery capacity. Favoring high capacity without sacrificing power, an 8S 3300 mAh and a 6S 4500 mAh propulsion battery were considered.

Equation X was used to confirm that the batteries did not exceed the maximum allowable energy storage of 100 Wh. The results yielded 97.68 Wh and 100 Wh for the 8S and 6S batteries, respectively. The 8S battery draws less current at higher cruise speeds, offering better efficiency by reducing I^2R losses. This is shown in figure X, where a higher motor input power correlates to a faster cruise speed. Flying at a higher speed allows for more laps to be flown in the five-minute flight window, which is advantageous for M2 and M3 scoring. Therefore, the 8S 3300 mAh battery was selected.



$$E = \text{Capacity (Ah)} * \text{Voltage(V)} * \text{Cell Count (S)} \leq 100\text{Wh}$$

Equation X

4.6.2 Motor and Propeller Sizing

The goal of the powertrain maximizes the dynamic thrust, cruise speed, and endurance while minimizing weight. Due to no takeoff distance requirement, static T/W ratio was not a significant factor in the sizing. Based on scoring analysis, M2 target cruise speed is 85ft/s, and for M3, the airplane must reach a cruise speed of 75 ft/s while towing the banner. An in-house optimization tool named PROM was constructed using Python to select the desired propulsion system in tangent with eCalc. PROM utilized BEMT, which uses differential equations along the propeller blade to calculate thrust at any throttle percentage, velocity, and battery state. The tool analyzed ~17,000 configurations, only considering 200-700 kV motors and propeller dimensions ranging from 7-22" diameter and 4-16" pitch. The primary constraints for the propulsion system were cruise speed as well as weight efficiency. PROM determined that a Scorpion II-4025 330kV motor with a 15x10 propeller achieved the best combination of cruise speed and weight for the five-minute flight window. The propeller choice was also verified with the mission requirement to make sure it is suitable for a moderate T/W ratio and a high speed powertrain. Using the eCalc tool, the selected propulsion configuration is summarized in Table X, together with the predicted current draw and flight time for both missions. The predicted max current remains comfortably below the 100 A limit, and the estimated flight time makes effective use of the available 5-minute window.

Table X. Final Propulsion System Configurations

Final Propulsion System Configurations		
Component	M2	M3
Motor	Scorpion SII-4025-330KV	
ESC	Phoenix Edge 75A	
Battery	Gaoneng 8S 3300mAh 100C	
Propeller (in x in)	15x10	
Total Weight lbs (kg)	2.5 (1.13)	
Max Continuous Current (A)	51	51
Cruise Speed ft/s	100	80
Endurance (minutes)	4.8	4.8

Thrust analysis was conducted to verify that the selected propulsion system would maintain the desired airspeed. The results demonstrate that the chosen configuration provides adequate thrust, achieving the minimum target airspeed for both missions comfortably. The thrust curve for the propulsion system was generated by PROM at 80% throttle as a factor of safety to meet the optimal cruise speed. There are two drag curves for M2 and M3 due to the increase in drag coefficient introduced by the banner, as shown in figure Y. The cruise speed for each mission was determined by the intersection between the thrust and the drag curve, indicated by the marked point in figure Y.

The maximum flight time can be calculated based on predicted current draw using equation Y, which assumes 85% of the battery capacity is discharged (2805 mAh) and 80% throttle is used, revealing a flight time of ~4.84 minutes, which meets mission objectives.

$$Time = \frac{Battery\ Capacity\ (Ah) * 0.85}{Current\ Draw} \times 60 \quad \text{equation Y}$$

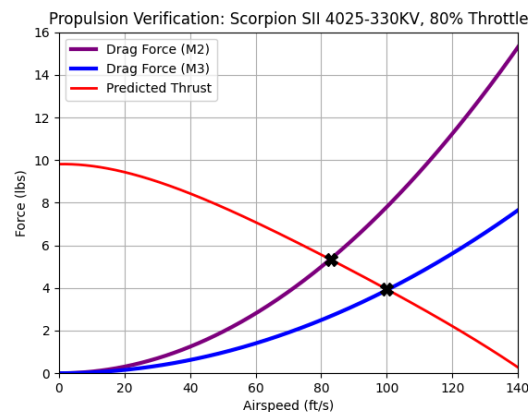
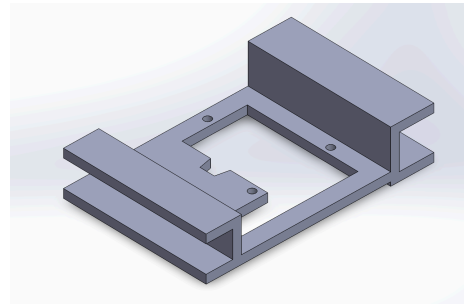


Figure Y

4.7. Avionics Sizing and Integration

The receiver and servos were selected based on several factors, yielding the Futaba R7008SB receiver, DS115CLHV servo for control surfaces, and DSServo 20kg servo for landing gear and banner release. The receiver chosen allows for eight channel inputs, accommodating propulsion, avionics, and banner systems. The smaller servos were assigned upon consideration of the force required to move the control surfaces, while stronger servos were assigned to the landing gear and banner deploy mechanism to handle heavier forces. To integrate the servos across control surfaces, the team designed 3D printed servo mounts specifically sized to fit between H-stab spars, the most structurally stable part of the airframe.



4.7 Stability Analysis

The static and dynamic stability analyses were conducted in XFLR5, with the former being verified with CFD analysis. Static and dynamic stability varied only slightly between each mission as varying amounts of ballast were used to maintain the same CG location. The changes in inertia between M1-M3 were analyzed using the dynamic stability analysis tool in XFLR5 to confirm the stability characteristics are still satisfied.

4.7.1 Static Stability

A static margin target of 8%, referencing previous aircraft designs, was set to allow the aircraft to have a high amount of maneuverability. Additionally, target values for static stability were chosen based on reference values from a Cessna 172. To ensure static stability in all flight regimes, the aircraft's center of gravity was strictly limited to between 20-35% of the wing chord. Thus, given a 17 inch root chord, the optimal center of gravity placement was found to be at 4.5 inches behind the leading edge, or at 26.5% of chord. Table ___ summarizes the aircraft's longitudinal, lateral, and dynamic stability for all three missions from XFLR5.

Static Stability			
Parameter	C_{m_α}	C_{l_β}	C_{n_β}
M1	-0.5834	-0.0245	0.114
M2	-0.5838	-0.0237	0.114
M3 (Banner Deployed)	-0.5838	-0.0245	0.114
M3 (Banner Released)	-0.7166	-0.0244	0.1214

A negative C_{m_α} value, a negative C_{l_β} value, and a positive C_{n_β} value corresponded to stable flight performance. The aircraft passed these basic static stability requirements. The values were also compared to those of a Cessna 172 to confirm they were within an ideal range. Once static stability was validated in XFLR5, a CAD model of the aircraft was analyzed in STAR-CCM+ to confirm values. The team found that the lateral and directional stability didn't change significantly, with a C_{l_β} of -0.033 [1/rad], and a C_{n_β} of 0.125 [1/rad]. However, the model had a significant difference in longitudinal stability, as the CFD simulation was able to more accurately analyze fuselage effects. The result showed a C_{m_α} value of -0.293 [1/rad], which corresponds to a static margin of 8.43% – very close to the target of 8%. The CFD simulations also showed that a -2° incidence angle on the horizontal stabilizer was necessary to trim the aircraft, and was thus implemented.

4.7.2 Dynamic Stability

The dynamic stability of the aircraft was analyzed using XFLR5's stability analysis module. Inertial estimates for the aircraft were obtained from wing, tail, and fuselage densities from the TAMU DBF 2025 aircraft as well as this year's MK1 aircraft. These inertial estimates were approximated using point masses in XFLR to estimate dynamic stability characteristics. The point mass distribution for M2 is presented in figure X.

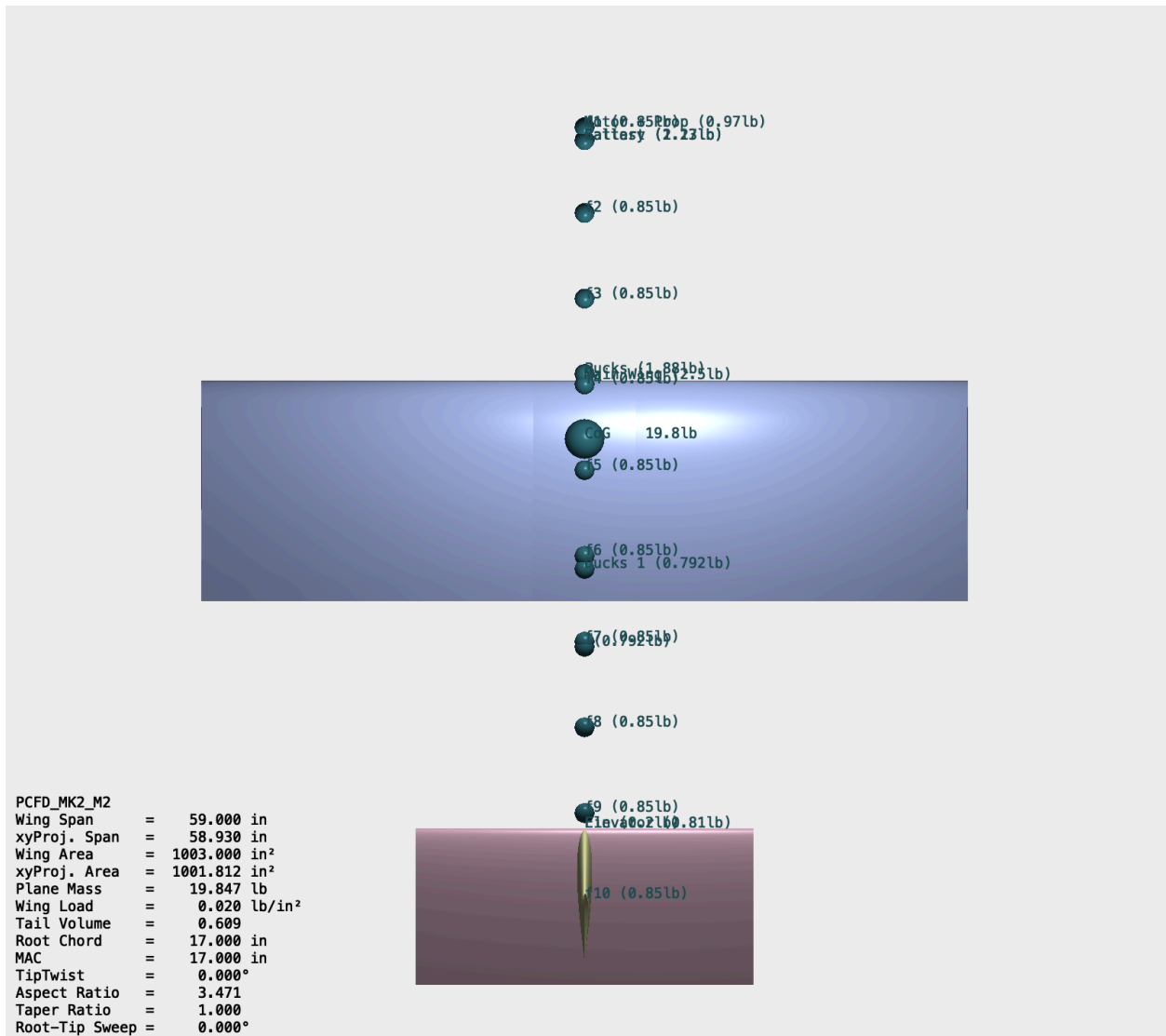


Figure X

The aircraft was analyzed for four different scenarios: M1, M2, M3 with banner deployed, and M3 with banner released. The shift in CG caused by releasing the banner was accounted for by placing the center of gravity without the banner further forward. Without the banner, the CG was placed 0.5 inches forward of the M1 and M2 position using ballast, causing the aircraft to be more stable. Once the inertia information was applied, the aircraft's short period, phugoid, roll, dutch roll, and spiral modes were analyzed. The dynamic stability of the aircraft was visualized using complex-plane eigenvalue plots, and are presented in figure Y and figure Z.

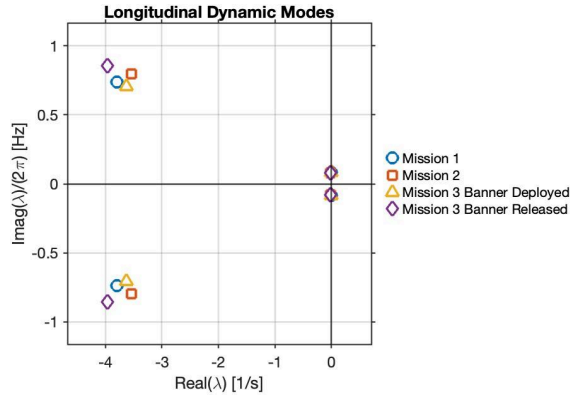


Figure Y

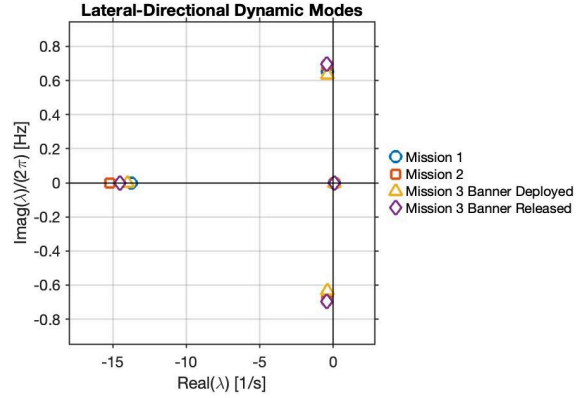


Figure Z

The longitudinal plot shows two stable second order modes, showing stability in phugoid and short period modes. The lateral-directional plot shows one stable 1st order mode, one stable 2nd order mode, and one unstable 1st order mode, showing that roll and dutch roll are stable, while spiral is unstable. The dynamic stability for each mission was further analyzed by comparing results to the flying qualities presented in MIL-F-8785C. The requirements for each level are presented below in [table __](#).

MIL-F-8785C Category B Requirements (L1/L2/L3)

MIL-F-8785C Category B Requirements (L1/L2/L3)					
Mode	ζ	ω_n	$\zeta \cdot \omega_n$	t_2	τ
Short Period	$0.30/0.20/0.15 \leq \zeta \leq 1.30$	—	—	—	—
Phugoid	$\zeta \geq 0.04/0.00/-$	—	—	—	—
Roll	—	—	—	—	$\tau < 1.4/3/10$
Dutch Roll	$\zeta \geq 0.08/-/-$	$\omega_n \geq 0.4/-/-$	$\zeta \omega_n \geq 0.35/-/-$	—	—
Spiral	—	—	—	$t_2 \geq 20/8/4$	—

These requirements were applied to the eigenvalues found from XFLR5 for each flight configuration and are presented in [table __](#).

Dynamic Stability Values (M1 / M2 / M3-Dep / M3-Rel)

Dynamic Stability Values (M1 / M2 / M3-Dep / M3-Rel)					
Mode	ζ	Ω_n (rad/s)	$\zeta \cdot \omega_n$ (rad/s)	t_2 (s)	τ (s)
Short Period	0.634 / 0.579 / 0.632 / 0.593	6.000 / 6.127 / 5.744 / 6.677	3.803 / 3.545 / 3.631 / 3.962	–	–
Phugoid	0.010 / 0.015 / 0.008 / 0.016	0.521 / 0.487 / 0.515 / 0.512	0.005 / 0.007 / 0.004 / 0.008	–	–
Roll	–	–	–	–	0.073 / 0.066 / 0.071 / 0.069
Dutch Roll	0.100 / 0.095 / 0.096 / 0.101	4.115 / 4.354 / 3.997 / 4.390	0.411 / 0.416 / 0.385 / 0.443	–	–
Spiral	–	–	–	9.87 / 10.45 / 10.12 / 11.16	–

The results are translated into flying quality levels as prescribed by MIL-F-8785C in [table ___](#).

Achieved Flying Qualities Level (per mission)

Achieved Flying Qualities Level				
Mission	M1	M2	M3 - Deployed	M3 - Released
Short Period	L1	L1	L1	L1
Phugoid	L2	L2	L2	L2
Roll	L1	L1	L1	L1
Dutch Roll	L1	L1	L1	L1
Spiral	L2	L2	L2	L2

As can be seen in the above table, XFLR5 estimates a level 1 flying quality in all dynamic modes except in phugoid and spiral modes. Based on previous TAMU DBF aircraft, it is known that instability in the spiral mode can easily be overcome by pilot input. It is also known that XFLR5 generally underestimates

drag significantly, which is a significant factor in the phugoid calculation. Through additional analysis using CFD, it was confirmed that the aircraft's drag was much higher than predicted by XFLR5, meaning that the aircraft's phugoid mode is much closer to a level 1 flying quality. Based on these additional factors, it was decided that the aircraft was stable and ready for the CFD analysis.

4.8 CFD Analysis

4.8.1 CFD Setup

The final aircraft model was analyzed using STAR-CCM+, a CFD program commonly used to compare and validate lift-curve trends, refine drag predictions, and identify areas of flow separation. The CFD geometry included the full aircraft shell (wing, fuselage, empennage, landing gear) with no control surfaces. The propeller was omitted to reduce meshing complexity.

The analysis consisted of an α sweep from 0° to 16° in 1° increments and a β sweep from 0° to 30° using finer increments at values close to 0. The aircraft was simulated in a clean configuration at 85 ft/s under atmospheric conditions accurate to Wichita, Kansas (ρ and μ set accordingly) with a Re of 8.8×10^6 based on the MAC. The simulations were performed using a steady state RANS solver with the k- ω SST two equation turbulence model. This method was chosen as it offers more reliability with adverse pressure gradients and separation.

The computational domain encompassed 50 chord lengths in both the positive and negative x, y, and z directions. This ensured enough distance to properly solve the wake. The inlet boundary was modeled as a velocity inlet with freestream velocity components set to the desired α and β . The exterior domain boundaries were modeled as freestream boundaries and represented the freestream atmospheric conditions. A global base mesh size of 1 ft 5 in was selected as it provided a conservative estimate of drag and a desired iteration count for convergence. A three-level mesh sensitivity study was performed by varying the global base mesh size $\pm 30\%$ relative to the baseline mesh. Local refinements on the wings, fuselage, and wake were employed and scaled off of the global base size to remain consistent. Each case was iterated until the residuals dropped at least three orders of magnitude and stabilized. The mesh statistics can be found in **TABLE X**. These mesh settings resulted in a total cell count of just over 9 million.

Final Mesh Statistics				
Parameter	Global	Wings	Fuselage	Freestream
Base Size	1 ft 5 in	.15 in	.25 in	8 ft
Growth Rate	1.1	1.1	1.1	1.1
Wake Spread Angle	-	10°	10°	-
Wake Refinement Distance	-	17 ft	17 ft	-

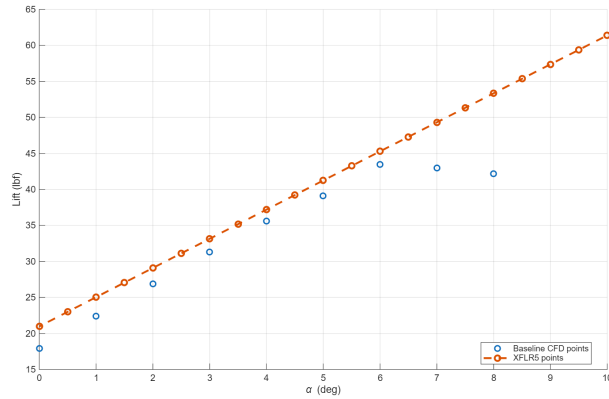
With the baseline CFD setup established, the team performed configuration trade studies. The intent of these studies was to isolate a few high impact geometry features and quantify how each one influences the aerodynamic trends. Each concept was evaluated using the same solver settings, meshing strategy, and boundary conditions described previously.

4.8.2 Trade Studies

Three targeted studies were performed: nose shaping, wingtip plate, and a rear strake. These studies were meant to assess the sensitivity of separation behavior, tip vortex management effects, and wake flow manipulation. Each configuration was analyzed for its aerodynamic effects and the complexity it would add to the aircraft. Due to limited computational resources, only a few data points could be obtained for each trade study. The remainder of the graphs were filled with a line of best fit.

4.8.3 Nose Cone Study

The first of these studies was the examination of the nose cone shape. At higher alpha values ($\alpha \geq 7^\circ$), unsteady flow separation originated off the top corners of the nose. This effect was significant as the flow separation from the nose washed over the root of the wing and horizontal tail causing an early onset stall near $\alpha = 7^\circ$. At $\alpha = 8^\circ$.



Simcenter STAR-CCM+

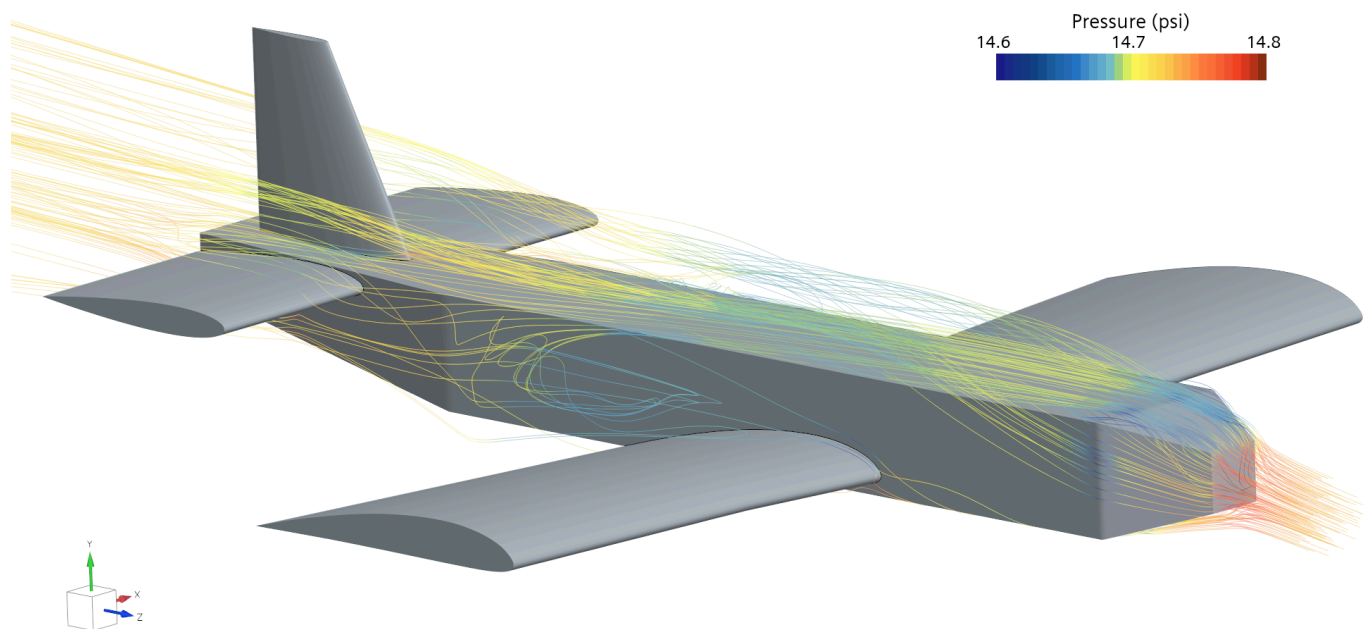


Figure X

Figure X shows the lift vs α graph compared to the XFLR5 analysis as well as the streamlines flowing over the nose at $\alpha=8^\circ$. An alternative nose was made with a rounded geometry to analyze the relationship between added curvature and stall angle. When using the same streamline definition from the baseline case, a noticeable decrease in the amount of flow separation was observed, shown in figure Y. This is significant as it reduces the separation over the wings, preventing an early stall. The results of this study confirmed the selection of the rounded nose cone for MK2.

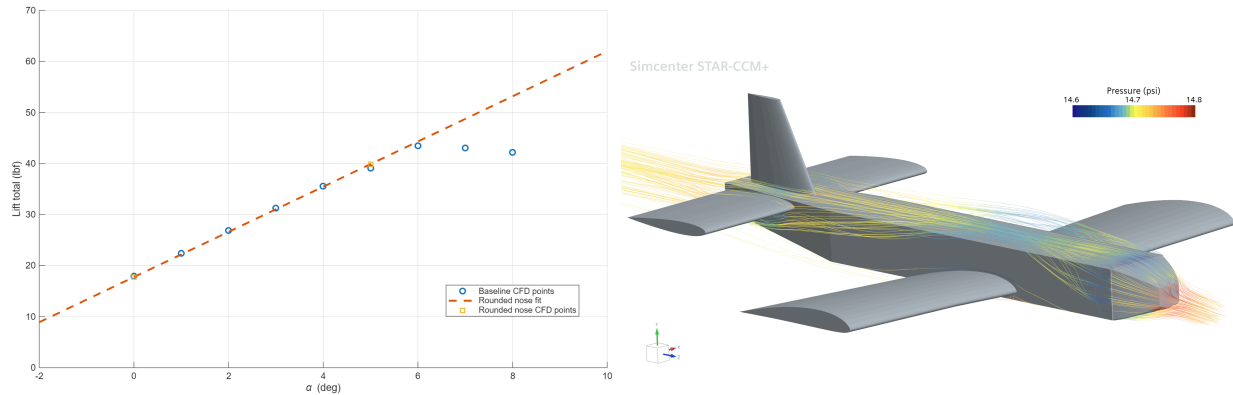


Figure Y

4.8.4 Wingtip Study

A second study was performed on wingtip devices in the form of an airfoil shaped plate as shown in figure X. The goal was to examine the aerodynamic benefits of the tip plate in areas like wingtip vortex strength and the effect on lift at limited alphas. The vortex strength was analyzed by looking at the effects on induced drag. When comparing the drag at $\alpha = 0^\circ$ to $\alpha = 8^\circ$, there was a 5% increase in drag and an 8% increase in lift as shown by the lines of best fit in figure Y. The vortex footprint can be seen in figures H and Z where an isosurface of Q-criterion (Q) is shown. Figure Z shows an isosurface of Q-criterion (Q) which can be interpreted as the difference between the local rotation and the local strain, where $Q > 0$ indicates the rotation is dominating the strain. This parameter is useful in highlighting the vortices shedding from the wingtip. The two figures highlight areas where flow separation is decreased near the tip, improving the lift distribution along the span. This can also be seen in the scalar coloring of the surface based on pressure where the lower pressure on the top surface indicates more suction based lift. The increase in drag was not favorable over the slightly larger increase in lift as lap time is a significant factor for optimal scoring in M2 and M3.

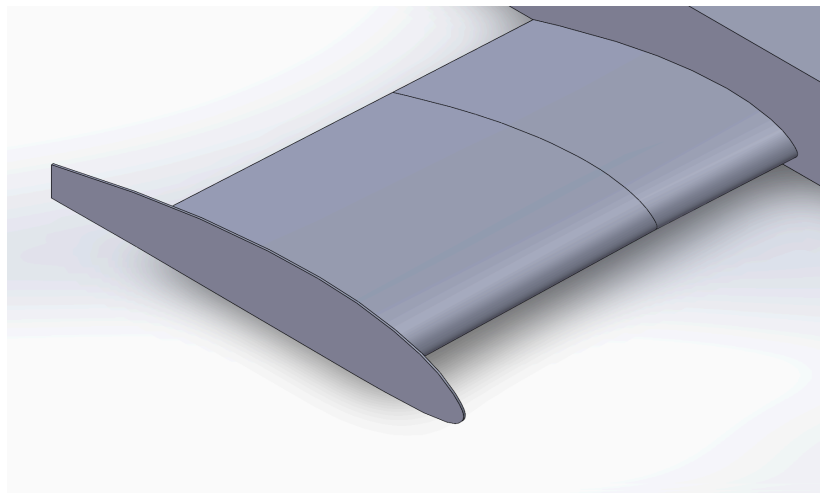


Figure X

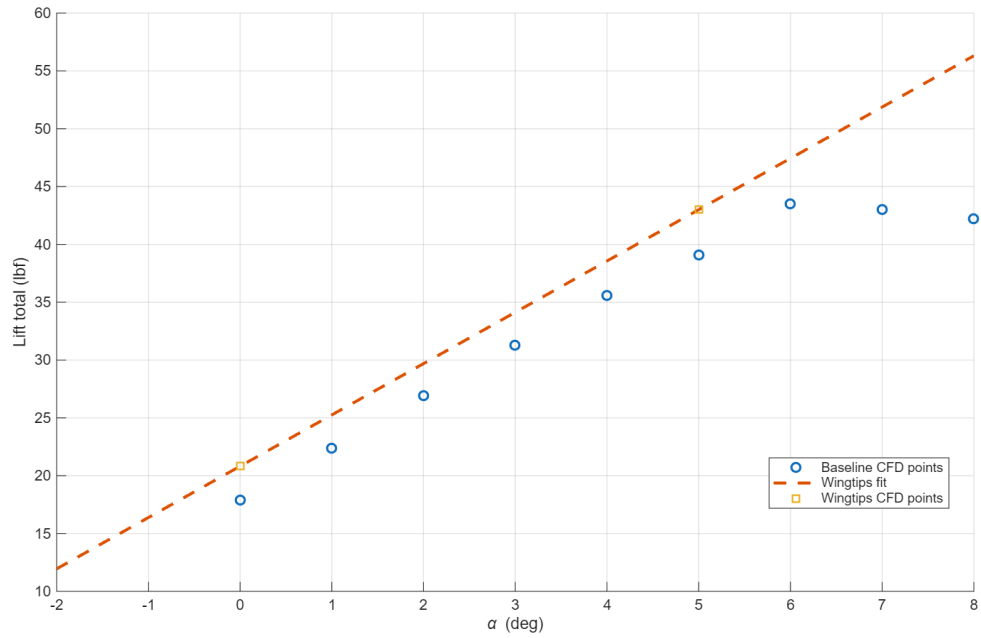


Figure Y

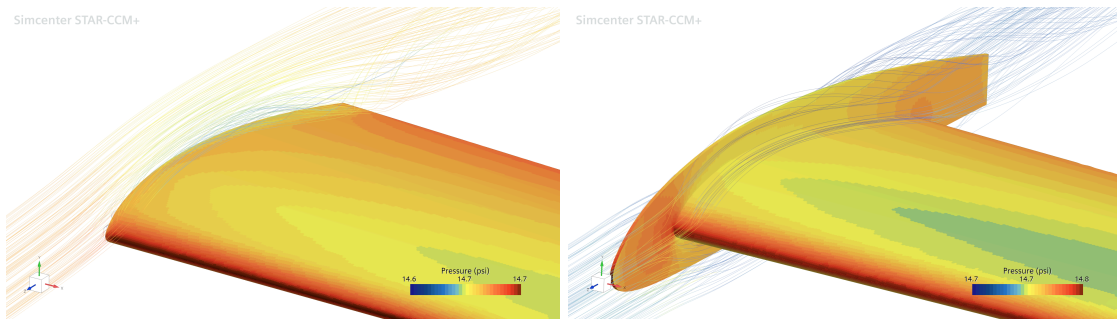


Figure H

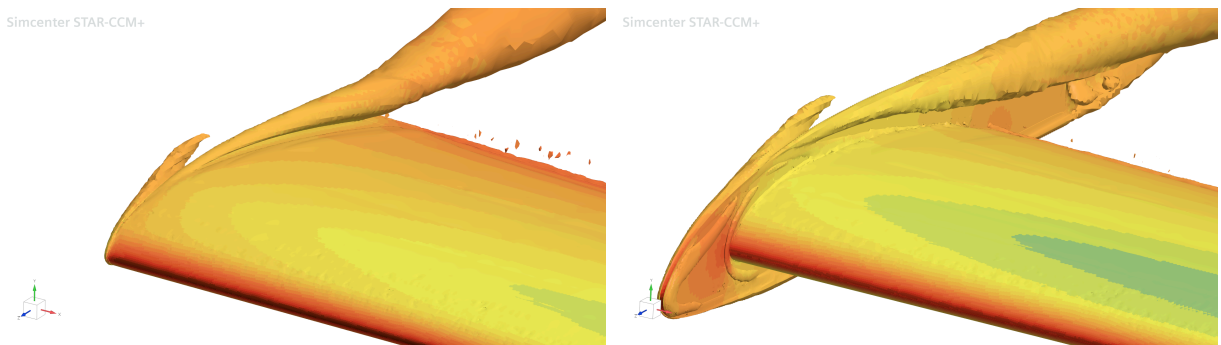


Figure Z

4.8.5 Empennage Study

The final study was done in an attempt to analyze wake manipulation devices like a strake on the underside of the empennage, as seen below in figure X. This small ventral strake was meant to address

the low energy, separated, and unsteady flow of the wake that would go on to interact with the banner. This unsteadiness can translate into banner oscillations and a high likelihood of the banner wandering side to side, creating de-stabilizing side forces on the aircraft. The addition of the ventral strake was intended to generate controlled streamwise vorticity that would produce a more centered wake behind the aircraft. The effect of this modification was analyzed with an isosurface of Q with the cleanliness of the wake being seen as reduced or more coherent Q structures. It was found that the effects of the strake did not provide sufficient benefit to overcome its potential negative interference during missions, such as a tail strike or during failure in banner deployment.

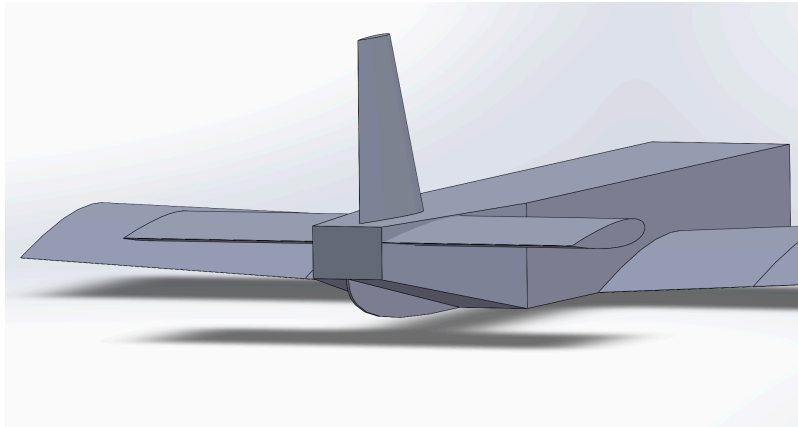


Figure X

4.9 Preliminary Performance Prediction

4.9.1 Preliminary Performance

From the results of the preliminary analyses, a variety of performance predictions were obtained and are displayed below in **TABLE X**.

Table X Predicted Preliminary Performance

Predicted Preliminary Performance			
Parameter	M1	M2	M3
Aircraft Weight (lbs)	15.93	19.85	16.48
Payload Weight (lbs)	-	4.95	0.19
Banner Length (ft)	-	-	5.83
Cruise Speed (ft/s)	85	85	75
Turn Speed (ft/s)	50	50	50

Lap Time (s/lap)	60	60	75
Number of Laps	3	5	4
Mission Time (sec)	180	300	300

4.9.2 Preliminary Performance Uncertainties

These preliminary performance predictions are strongly dependent on simplifying assumptions and limited validation data. Lap counts and lap times assume constant straight-line and turn speeds, with an idealized turn radius based solely on a fixed bank angle and no energy loss in the turns. In reality, turn performance will vary based on thrust, produced lift, control authority, and the actions of the pilot. Mission weights are built from estimated subsystem and payload weights and do not include potential growths from repairs, wiring, etc. For Mission 3, banner length and banner drag are computed using an assumed linear drag coefficient vs. Reynold's number relationship, derived from external reference data. Actual banner drag can vary significantly with fabric type, edge treatment, flutter, twist, and deployment behavior. Lastly, atmospheric density, wind, and turbulence can change effective airspeed and lap time relative to these predictions. Because of these factors, the values in this table should be treated as order of magnitude estimates useful for early trade studies until validated by ground testing, flight testing, and banner drag testing.

5 Detailed Design

5.1 Dimension Design Parameters

Table X. Final Dimensional Design Parameters

Final Design Parameters					
Fuselage		Wing		Empennage	
Length	5.50 ft	Airfoil	N22	Airfoil	NACA 0012
		Span	4.92 ft	Horizontal Stabilizer Span	2.14 ft
Width	8.00 inch	Area	1003 inch ²	Horizontal Stabilizer Chord	1.08 ft
		Aspect Ratio	3.471	Horizontal Stabilizer Incidence	-2°

Height	5.75 inch	Dihedral Angle	+3°	Vertical Stabilizer Span	10 inch
		Incidence	+2° with a twist of +1.5°	Vertical Stabilizer Chord (Root & Tip)	5 inch / 10 inch

Final Design Parameters - Control Surfaces			
Surface	Span	Mean Chord	Max Deflection
Aileron	10.50 in	4.25 in	±45°
Elevator	8.25 in	4.80 in	±45°
Rudder	7.00 in	2.50 in	±30°

Final Design Parameters - Avionics					
Motor		Propellers		Controls and Power	
Model	Scorpion SII 4025-330	Manufacturer	APC propellers	Receiver	Futaba R7008SB
Voltage	330 kV	M1 (Diameter x Pitch)	15x10 in	Servos	DS115CLHV
Max. Power	2000 W	M2 (Diameter x Pitch)	15x10 in	Battery Model	Gaoneng Battery
Max. Operational Current	75 A	M3 (Diameter x Pitch)	15x10 in	Cell Count	8S
Weight	11.96 oz			Pack Voltage	33.6 V

5.2 Structural Analysis

Structural analysis using Ansys Mechanical, an FEA software, was conducted on the wings and fuselage of MK1 to determine wing loading and landing effects. Pressure maps of the wings were created to visualize the stress distribution across each rib and to confirm structural integrity of the wing spars under flight conditions. To simulate this analysis, surface load data obtained from the CFD simulations was mapped to each wing rib and truss structure on the fuselage as parasolids. A python script using data from the simulations was run to determine the loading positions along the aircraft, identifying high stress regions and potential points of failure. The analysis focused on determining the total deformation, normal stress, maximum principal stress, and directional deformation of the wings and fuselage under simulated flight conditions. Based on these results, MK1's design was iterated upon to improve MK2's structural performance.

5.3 Banner

An initial prototype banner was constructed out of plastic grocery bags, cardboard, and string during the design process for testing purposes. To test the banner, it was held by the far end of the tether and pulled at a set speed simulating an aircraft. To attach the banner to the tether and the tether to the plane, a process known as rigging was applied. The rigging design of the banner was inspired by full sized banner towing rigging. There are two points of contact at the top and bottom of the leading edge of the banner. These two points of contact then converge into one tether that is the final point of contact between the aircraft and the banner. Various rigging ratios were tested utilizing the prototype banner. The optimal ratio was achieved by rigging the bottom of the banner down $\frac{1}{5}$ of its height, seen in figure X. This offset allows the banner to fly below the aircraft's fuselage and keep the tether taut.

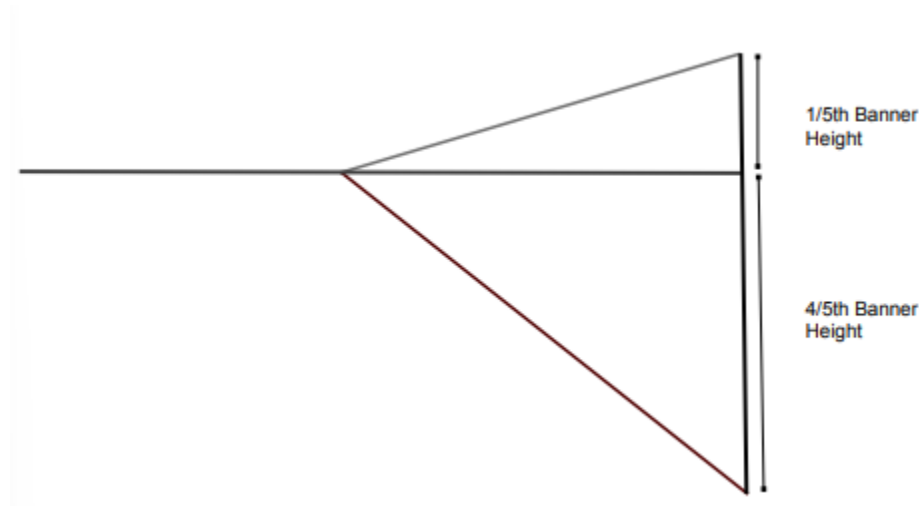


Figure X. Tether Rigging Ratios

In addition to the tether, the rigging consists of stainless steel grommets, kite twine, and split key rings to attach the tether and shock cord to the servo arm in the banner mechanism. The stainless steel grommets and kite twine were chosen for their durability, accessibility, and high strength. In order to minimize banner drag and size, as well as to optimize weight and M3 score, the aspect ratio chosen for the banner was 5. After testing the banner, the ideal length is 5.83 feet.

It was found during testing that to keep the banner vertical, weights would need to be added to its bottom edge. Washers were chosen due their low price, accessibility, and variety of masses. The washers were hand-sewn into the banner with a piece of fabric, creating a pocket to hold them. The chosen fabric, nylon ripstop, was found to have issues with fraying. To address this issue, the edges of the banner were hemmed. Regarding the logo, a custom-design was printed on heat transfer vinyl (HTV) and then ironed onto both sides of the banner. To anchor the kite twine, grommets were pressed into the banner using a press.

The banner is attached and released from the aircraft using a servo motor. A 3D printed banner release mechanism was designed to ensure that the servo arm goes through the loops of the tether and shock cord with ease (figure X). The first iteration of the release mechanism design was a rectangular block that had two rectangular slits for the loops and a horizontal through hole for the servo arm.

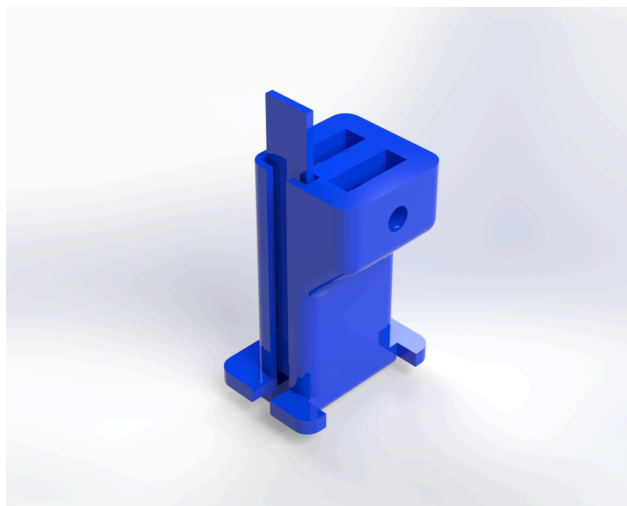


figure X. The first iteration of the banner mechanism part

Mounting the banner to the aircraft using the first design was found to be time consuming and inefficient during GMruns. To address this, a slider was added in the second iteration (figure Y) that can be removed during loading and then slid into the release mechanism to push the tether down, such that it does not get caught. Additionally, an O-ring was added to the end of the tether and shock cord, creating a stiffer loop for the servo arm to be threaded through. Another difficulty with the first design was keeping the string used to stow the banner underneath the plane taut, and a shock cord was implemented in the second iteration to address this.



figure Y. Final iteration of the banner mechanism part

To attach the banner release mechanism during the staging period prior to M3, a banner release plate that would be pushed into the fuselage with a velcro attachment was designed. The 3D printed plate holds the entire banner release system, including the servo, release mechanism, and shock cord hook. The servo is mounted with screws. The banner release mechanism and shock cord hook are both a part of the 3D printed plate. When mounting the banner release system into the fuselage, a thin 3D printed plate is popped out, and the banner release plate is pushed into its place with a friction fit.

5.4 Landing Gear Structural Analysis

An FEA study was conducted to validate the landing gear designs by loading the assembly models into Ansys Mechanical, for which fasteners had fixed constraints and axle contact points were subjected to loads representing a 2-3g landing impact. The simulation identified a peak Von-Mises stress of 52.86 MPa at the rear rod-to-mount interface, justifying the transition from PETG to PETG-CF to prevent fractures observed in previous iterations. A significant disparity between the stress experienced by the rear gear (52.86 MPa) and the front gear (6.24 MPa) confirms the design assumption that the rear gear bears the majority of landing forces. Furthermore, the stress maps reveal that the aluminum rods in the rear gear sustain most of the impact while remaining well within the material's factor of safety and elastic deformation range. While this data provided critical validation for the material transition, certain modeling approximations were made regarding the PETG-CF mounts to allow for a reasonable structural evaluation. Ultimately, these results confirm that the final iteration successfully handles the aircraft's weight and dynamic forces during high impact landing conditions.

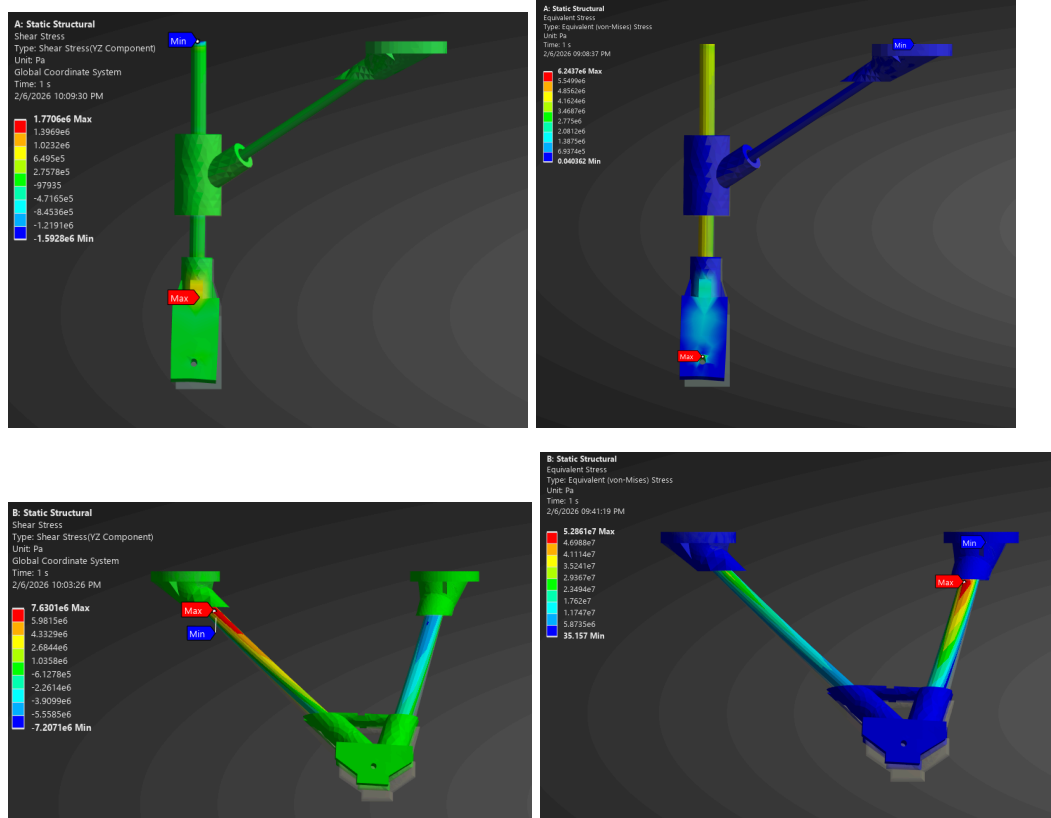


Figure X Landing gear FEA

5.4 Avionics and Wiring

The propulsion system is powered by the main battery which includes a 100A fuse for circuit protection, an electronic speed controller (ESC) that regulates power, a motor, and a propeller that generates thrust. The avionics system operates on a separate avionics battery which includes a switch that powers the receiver, distributing control signals to the aircraft's moving parts. The aircraft possesses two aileron servos, two elevator servos, one rudder servo, one banner deployment servo, and one front landing gear servo, as shown in figure X

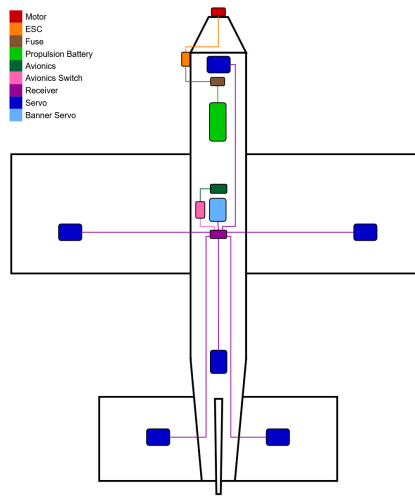


Figure X wiring diagram

The positioning of various avionics was based on a variety of factors emphasizing safety, access, and simplicity. The ESC placement was unusual due to the positioning of the cargo at the top of the fuselage. Because of the difficulty of running wires to the top of the fuselage and the danger associated with placement on the underside of the fuselage, the ESC was positioned on the side of the fuselage. This location ensures that the ESC receives proper cooling while also keeping it safe from potential hazards. In order to simplify wiring, the motor, fuse, and ESC were placed near the nose, maintaining a minimum of 8 inches between the propeller's plane of rotation and the fuselage, while the battery was implemented near the rear end of the fuselage. The battery's positioning was kept slightly variable in order to prioritize weight distribution and minimize the need for ballast. The avionics receiver, battery, and switch were subsequently placed centrally in the fuselage to allow for easy access and CG balance.

5.5 Weight and Balance

The aircraft's mass measurements were conducted in two phases. The first phase consisted of determining a preliminary mass through SolidWorks' Mass Properties tool, where materials were assigned to each component using a custom material library, allowing for the weight, inertial properties, and CG to be estimated. The custom material library was created through analyzing and recording wood properties from various databases. After completion of manufacturing and assembly, these estimates were then confirmed by measuring the reaction forces on all 3 of the aircraft's wheels using weight scales. An excel sheet takes the values of each scale and calculates the location of the CG using a sum of moments solution. Ballast weight and component placement are adjusted until the true CG is within a tolerance of ± 0.05 inch of the desired CG.

5.6 Flight Performance and Mission Specific Performance

The flight and mission performance parameters given in Table # were found to best optimize each mission's score based on the sensitivity analysis presented in Section 3. Aircraft weight, speed, and lap time was calculated using the finalized design parameters given in Section 5.1.

Table # Final Flight and mission performance parameters.

Flight and Mission Performance Parameters				
Parameter	M1	M2	M3	GM
Payload (lbs)	-	4.95	0.19	-
TOW (lbs)	15.93	19.85	16.48	-
Cruise Speed (ft/s)	85	85	75	-
Average Throttle (%)	80	80	80	-
Battery Usage (%)	30	85	85	-
Number of Laps	3	5	4	-
Mission Time (sec)	180	300	300	62
Estimated Score	1	1.849	2.849	0.1

5.7 Drawing Package

6 Manufacturing Plan

6.1 Manufacturing Process

The manufacturing processes included laser cutting, additive manufacturing, application of heat-shrink laminated cover film, and birch skinning. These processes were chosen due to their repairability and relatively low cost.

6.1.1 Laser Cutting

All of the custom wood components were created by laser cutting. Laser cutting uses concentrated amounts of infrared light to burn thin lines through a given material, making the cut extremely accurate with negligible material loss. The precise dimensional requirements of wing ribs and tight tolerances of fuselage panels make laser cutting the optimal tool for manufacturing these parts. It is also extremely

quick, cutting at speeds of up to 75 mm/s. This allows for rapid prototyping and a shorter manufacturing window.

6.1.2 Heat-Shrink Laminated Cover Film

Neucover and Doculam are lightweight heat-shrink laminated cover films used to create the smooth outer skin over the aircraft structure, enhancing aerodynamic efficiency. They also seal and protect the wood from moisture and are not prone to wrinkling. A hot iron is used to apply the film to a desired area on the plane. Next, a heat gun with optimal temperature and air flow settings is used to tighten and flatten the surface. These settings were used to avoid burning holes into the film or warping the wood.

Based on Table #, Neucover and Doculam are the most cost effective options available. Because Doculam is lighter and transparent, it is primarily used to reduce weight and allows the team to inspect the aircraft and analyze wiring and structural integrity without having to remove the film to do so. Neucover has better durability and adhesion. It is applied to wingtips and control surfaces to enhance pilot visibility. Additionally, Neucover is used for sections more prone to tearing and puncturing.

Skinning Material					
Criteria	Importance	Monokote	Neucover	Ultrakote	Doculam
Cost	5	1	4	2	4
Weight	5	2	2	1	3
Flexibility	2	1	3	3	2
Adhesion	3	2	2	4	1
Durability	1	4	2	2	1
Rigidity	2	4	2	2	1
Transparency	1	1	2	1	5
Total:		36	50	40	50

6.1.3 Birch Skinning

Birch skinning is a technique primarily used to reinforce the leading edge of lifting surfaces. This method was deemed to be more cost effective and time efficient than the alternative method of using composite leading edges. Birch skinning is done by cutting rectangles out of a 1/64th inch birch sheet and placing them inside a plastic bag. A steamer continuously fills the bag with steam and infuses into the birch. After 15 minutes, the birch is flexible and can fully conform to a leading edge. The birch is glued and allowed to fully dry.

6.1.4 Additive Manufacturing (3D printing)

3D printing was the preferred additive manufacturing method for specialized components with unique geometries, as they need to be rapidly iterated upon to fulfill complex functional requirements. These parts include the horizontal stabilizer spacers, servo and landing gear mounts, and banner release mechanism. The main downside of this method is the unpredictability of structural failure modes.

To address this, components that required structural integrity, such as the landing gear mounts, were tested within a factor of safety for compressive, shear, and cyclical loading, for which the final iterations passed.

6.2 Material Selection

A wide selection of materials were evaluated based on weight, cost, strength, and manufacturability, in order to determine the most optimal materials for construction. Among the considered materials were wood, carbon-fiber composites, plastics, foam, and metal.

6.2.1 Wood Selection

Wood was selected as the most suitable material for the majority of the aircraft due to its accessibility, low cost, and favorable strength-to-weight characteristics. A combination of poplar, pine, and balsa was ultimately chosen based on a weighted evaluation of their strength, weight, flexibility, and cost. Each component was assigned a material from the list depending on their structural role.

Poplar was deemed to be the preferred material for primary structural components such as ribs, bulkheads, and fuselage panels. Poplar offers significantly lower weight while maintaining adequate structural strength for load-bearing applications at a low cost. Although birch exhibits higher strength than poplar, its increased density and tendency to splinter when cutting make it less suitable for primary structure. However, 1/64 inch birch plywood does not share these issues, thus it is used as skinning material to provide localized torsional stiffness to leading and trailing edges of wings and stabilizers. Balsa was designated for elements where minimal weight is the primary requirement. Despite being neither particularly strong nor inexpensive, its extremely low density makes it effective for non-structural components, such as intermediate ribs, reducing the aircraft's overall weight.

For spars and longerons, wood types were evaluated separately due to their critical role in carrying bending loads. Basswood and pine were considered, with bending strength and weight prioritized over cost and flexibility. Pine achieved the best overall performance, offering a strong balance of structural capability, moderate weight, and workable flexibility. While basswood is both strong and inexpensive, its

excessive weight makes it unsuitable for spars in this design. Consequently, pine was selected for all wooden spars and longerons.

6.2.2 Plastic Selection

Multiple types of material were used for the various 3D printed components throughout the aircraft, with each material being chosen for particular structures by optimizing for weight, cost, and to ensure that each part would function as intended. The three materials selected were thermoplastic polyurethane (TPU), polyethylene terephthalate glycol-modified (PETG). TPU was used for the hatches and latches of the aircraft due to its rubber-like flexibility and durability. While the cost is high, the small number of hatches and latches allowed for its use. PETG was used the most generally for the printed components of the aircraft, due to its overall functionality and ability to be applied to a multitude of tasks, such as the banner system, servo mounts, and various connection pieces which all are required to be sturdy throughout flight. Finally, PETG-CF was used for the landing gear components, due to its higher strength than PETG and increased tensile strength, which is especially important for the landing gear which endures the most stress during landing.

6.2.3 Composite Spar Selection

Composite spars were preferred over wood for the wings due to their high strength-to-weight ratio. The four composites considered for this purpose are roll-wrapped carbon fiber, pultruded carbon fiber, Kevlar, and fiberglass. **Table X** displays the criteria that each composite was assessed on. Roll-wrapped carbon fiber performed best overall, due to its superior torsional strength, axial strength, structural rigidity, and low weight. There are two categories of roll-wrapped carbon fiber: unidirectional (UD) and multidirectional (MD). UD offers high stiffness under uniaxial loads while MD offers high torsional stiffness under shear loads. MD roll-wrapped carbon fiber rods were chosen for their high torsional stiffness, which is crucial for the shear loads generated in the wing spars.

Table X. Trade Study for Selected Composites Used For Spars

Trade Study for Composite Spar Material Selection					
Criteria	Importance	Roll-Wrapped Carbon Fiber	Pultruded Carbon Fiber	Kevlar Fiber	Fiberglass
Cost	4	2	3	4	4
Weight	4	3	3	1	1
Structural Rigidity	2	3	3	3	2
Torsional Strength	3	4	3	3	2

Axial Strength	3	4	3	3	2
Total		50	48	44	36

6.2.4 Adhesive System Selection

In order to bond and fix elements of the airframe together, various glues and epoxy are utilized. The chosen glue used was Cyanoacrylate (CA) glue due to its fast curing and setting time. For loose fitting, non-structural components, Maxi-Cure and Insta-Cure+ CA offered quicker drying times with minimal added weight. For structural components that required a larger amount of bonding strength, 5 Minute Quick-Cure Epoxy resin and hardener were used at a 1:1 ratio. For joints and components that required a moderate amount of strength, a 15 Minute Mid-Cure Epoxy was applied. Finally, for extremely high load bearing parts such as the wing spars and wingbox, Loctite EA 9640 provided the necessary strength to ensure any parts or joints would not separate.

6.2.5 Foam

Foam usage on the aircraft was limited to the passenger compartment. Foam was chosen as the material for this component because of its lightweight and rigid properties, as well as cost efficiency and accessibility. Two types of foam were implemented into the design, Kingspan Greenguard high density foam for the lower passenger board and a low density foam (1 inch thick GoTo Foam sheet) for the upper passenger cover. The lower passenger board required a firmer foam to allow for slot placement of the passengers while securing them. The upper passenger cover utilized softer foam to reduce weight while maintaining the positions during maneuvers.

6.2.6 Banner System Materials

When selecting the banner material, four main factors were considered: weight, weave characteristics, durability, and the fabric's ability to display a logo. To reduce the shift in CG after banner deployment, it was crucial to choose a lightweight material. For the banner to fly vertically behind the aircraft, air must flow through the weave of the banner fabric. Adding holes increased airflow, but negatively affected the durability of the banner. The best rule-compliant fabrics were found to be polyester or nylon ripstop. Both nylon and polyester ripstop are lightweight, durable, and flexible. Additionally, the ripstop design still allowed for airflow through the fabric. Nylon ripstop was ultimately chosen because it had a higher strength-to-weight ratio than polyester. To display the logo on the banner, a custom heat transfer vinyl was used. This method allows for application and curing on the Nylon ripstop fabric.

Table X. Design Matrix for choosing the best material for the Banner.

Banner Material Decision Matrix
--

Criteria	Importance	Vinyl	Nylon Ripstop	Polyester Ripstop
Weave Characteristics	5	1	5	3
Weight	4	1	4	4
Durability	3	5	4	3
Total		7	13	10

The tether for the banner was created out of kite twine. Kite twine was chosen because it is light, flexible, and designed for towing. To stow the banner beneath the fuselage, a marine-grade shock cord was chosen for its elasticity, resistance to cyclic loading, and low weight. In contrast, non-specialized elastics are prone to plastic deformation over time and could fail in flight. To keep the banner vertical, washers will be used as weights due to their low price, accessibility, and variety of masses. Stainless steel grommets were chosen for the rigging system due to their high tensile strength.

3D printing was selected to manufacture the release mechanism due to its complex geometry and the need for a durable, lightweight component. Polycarbonate (PC) was used to provide a rigid structure. Compared to other filaments, PC is relatively inelastic, which is critical for components that will be under consistently high loads when holding the banner. Lower-temperature thermoplastics may soften and deform under outdoor conditions, making PC the preferred choice for the banner release mechanism. Metal O-rings were used to secure the servo pin instead of tying loops on the tether and shock cord. This is because looping the tether and shock cord made it difficult to set up the banner mechanism during the loading process, as the loops tend to fall limp and miss the servo. The O-rings are important as they speed up the banner loading process for GM.

6.3 Assembly Process

Assembly processes were decided on factors including weight, manufacturing timeliness, and material selection.

6.3.1 Fuselage (Nose, Fuse, Wingbox)

The fuselage was constructed using $\frac{1}{8}$ inch poplar plywood sheets and $\frac{3}{8}$ inch square longerons spanning across the non-tapered body. The poplar sheets were laser cut to the geometries determined by the CAD model, and the longerons were cut to length using a jigsaw. The exterior walls of the fuselage were designed with mechanically locking geometries, allowing them to be fixed in position and bonded with 5 Minute Quick-Cure Epoxy. The longerons were laid along each inner corner of the rectangular fuselage shell and then bonded with 15 Minute Mid-Cure Epoxy. The nose, motor mount, and wingbox were constructed from multiple layers of poplar plates bonded together with LOCTITE EA 9460. The exterior of the fuselage was sealed against air flow using heat-shrink laminating cover film. The hatches

composing the upper fuselage shell were attached using 3D printed TPU latches. The nose hatch was fabricated from PETG, then friction fit and secured with painters tape as a secondary measure.

6.3.2 Empennage (V-Stab, H-Stab, Tailbox) (Tate)

The empennage consists of the horizontal and vertical stabilizer (H-stab & V-stab), and the tailbox which mounts them to the fuselage. The V-stab and each half of the H-stab were assembled separately using jigs to ensure proper spacing and alignment of the ribs. Each rib was laser cut out of either $\frac{1}{8}$ inch poplar ply-wood or balsa sheet depending on their structural role. The structural ribs comprising the root and tip rib of each component, the servo mounting ribs, and the outermost ribs of the control surfaces utilized poplar for its high strength. The intermediate, non-structural ribs were laser cut out of balsa sheet to save weight while maintaining aerodynamic shape. When assembled using jigs, CA glue was used to attach the ribs to the $\frac{1}{4}$ and $\frac{3}{8}$ inch width square pine spars. Furthermore, $\frac{1}{4}$ inch pine dowels were used for the control surfaces. For the leading edges, birch plywood capspar were inserted above and below the pine spars along the ribs to allow the skinned leading edges to be attached using CA glue. $\frac{1}{64}$ inch birch plywood skins were glued to the top and bottom of the ribs' trailing edges with CA glue as well. These skin acted as mounting surfaces for lamination.

The tailbox was assembled using two side bulkheads that would provide a point of contact for both H-stabs. The two H-stab spars were inserted into the bulkheads on either side, with an additional point of contact from the external fuselage shell. Additionally, a top bulkhead was connected between the two side bulkheads to provide a mount for the V-stab spars, which also used the fuselage shell as secondary point of contact. To mount the tailbox within the airframe, a front and rear bulkhead, also mechanically joined with the side and top bulkheads, were inserted into slots in the fuselage. Because the tailbox is a medium load bearing component housing the H-stabs and V-stab, 15 Minute Mid-Cure Epoxy was the preferred adhesive to secure the interconnected parts, without adding excessive weight and cost of using Loctite EA 9460.

6.3.3 Wings (Kody)

For the wings, there were three categories of rib. The first were the base ribs, which were the normal N22 airfoil with trussing for weight reduction and the necessary holes for mounting to the spars and jigs. Next were the control ribs, which were like the base rib except the trailing 4.5 inches were cut off to make room for the ailerons. Two of the control ribs also served as mounting points for the servo mount, which was a 3D printed part that slotted into the trussing of the control ribs and was glued in place. This mount had a cut out and screw holes for securing the servo. Last were the transition ribs. These were placed next to the outer two control ribs and were like the base rib but had an extra hole that would hold the aileron dowel.

The wing assembly was nearly identical to the H-stab. Each side of the wings were assembled using laser cut ribs made out of $\frac{1}{8}$ inch poplar or balsa depending on the structural importance of the rib. The ribs

were then mounted on the wing jigs to ensure correct spacing and incidence. The wings used a square carbon fiber tube for the main spar and a circular carbon fiber tube for the secondary spar, which were both attached to the ribs using 15 Minute Mid-Cure Epoxy. The trailing edge pieces $\frac{1}{2}$ inch wide strips of $\frac{1}{64}$ inch birch that were glued using CA to the top and bottom of the ribs trailing edge. These pieces helped keep the ribs from bending and also provided a large flat surface for lamination. The leading edge consisted of a large sheet of $\frac{1}{64}$ inch birch and two $\frac{1}{8}$ inch poplar plates that were glued into slots cut into the top and bottom of each rib. The birch sheet was first steamed, and once ready was bent around the leading edge and glued to the two poplar plates. A balsa plate was also added between the two root-most ribs to prevent the root rib from being bent inward by the laminated cover film.

6.3.4 Control Surfaces (Karl)

The rudder, elevators, and ailerons were laser cut out of $\frac{1}{8}$ inch poplar and constructed separately before being attached to their aerodynamic surfaces. To align each control surface's ribs, a rectangular $\frac{1}{8}$ inch poplar plate was inserted through them. After rib alignment, a $\frac{1}{4}$ inch wooden dowel was inserted through the quarter chord of each rib to enable actuation. The dowels were glued with CA to each rib, but not to the tip and root ribs of the stabilizers and wings to allow for free rotation. To keep the dowels in their set position, wooden spacers were placed at the ends of the control surface ribs. $\frac{1}{64}$ " birch strips were attached to all TE edges to allow for smooth application of the Doculam heat-shrink film. The control surfaces were then attached to the wings, H-stab, or V-stab by inserting the dowels into the ribs next to control surface end ribs. A control horn was attached between the servo ribs to the rectangular plate using CA glue for each control surface.

6.3.5 Exterior Lamination (Kody)

Doculam and Neucover films were applied as a final laminating skin over the entire plane to fill holes in geometry from trussing and gaps between ribs. Smoothing irons provided the heat needed to activate the adhesive layer and the pressure needed to ensure the film seals to the material. Due to the large amount of empty space between ribs, the birch skins on the leading and trailing edges act as anchors for the film to fully adhere to. This applies to the wings, H-stab, and V-stab, ensuring the film does not unbind while a heat gun is used to tighten the film. Heating the film removes wrinkles and ensures the desired aerodynamic shape over the entire body of the plane.

6.3.6 Payload & Mission Components

The M2 payload included two separate components, the passenger compartment and cargo container. Both the cargo container and structural shell of the passenger compartment are composed of laser cut $\frac{1}{8}$ inch poplar plates. The plates are fixed into position via mechanical locking features and then bonded with 5 Minute Quick-Cure epoxy. To secure these frames to the aircraft during a mission, a 3D-printed TPU latch is screwed onto the side of the frames and latched to the side of the fuselage. The passenger

compartment's foam boards were cut from sheets into the required rectangular profile with circular slots. The foam boards were attached to the frame via painters tape.

The M3 payload included two separate components, the banner and the release mechanism. The release mechanism was entirely 3D printed PC, and relied on a **SERVO FOR RELEASE**. The release mechanism is connected via nuts and bolts to the bottom of the fuselage. During stowage, the banner is strapped to the bottom of the fuselage with a marine-grade shock cord looped into the release mechanism. The tow line of the banner is a kite string that was cut to length with scissors and then connected to the banner by looping through the banner's stainless steel grommets. The grommets were installed using a **Grommet tool**. The banner itself is a sheet of nylon ripstop cut to shape with pinking shears to reduce fraying. Additional fabric was sewn in a rectangular shape around the leading edge of the banner.

6.3.7 Landing Gear (Uzair)

The assembly process of the landing gear was conducted through phases of testing, and then integration into the aircraft. The design of the landing gear and its orientation were previously done in SolidWorks and used as a guide during assembly. A rectangular wooden block was used as a test rig to mimic a fuselage panel that the landing gear would attach to for various tests. The aluminum rods were set to their appropriate length and then slotted into their respective landing gear mounts. The collars and wheel mounts were slid onto the rod from the ends and placed in their respective locations. To ensure the aluminum rod stayed in position and the mounts wouldn't detach, they were taped together for the test. However, LOCTITE EA 9460 was used to seal the rod and the mounts together for the finalized design after testing was completed. The landing gear components were bolted onto the bottom of the testing rig and then various tests such as cycle testing, stress testing, and impact testing were performed to ensure functional requirements were met. Once the design was validated through the testing phase, the landing gear was then bolted onto the actual fuselage.

6.4 Manufacturing Timeline (Pierre)

Manufacturing Plan January 1st to April 16	January				February				March				April		
	Wk.1	Wk.2	Wk.3	Wk.4	Wk.1	Wk.2	Wk.3	Wk.4	Wk.1	Wk.2	Wk.3	Wk.4	Wk.1	Wk.2	Wk.3
Design & Analysis															
CAD															
Manufacturing Mk. 2	Build to Flight Test on 1st Week of March ★														
Material Procurement															
Laser Cutting															
Jigs															
Wings															
H-Stab															
V-Stab															
Control Surfaces															
Birch Skinning															
Fuse															
Nose															
Wingbox															
Tailbox															
Laminating															
Landing Gear															
Banner Mechanism															
Payload Mounting System															
Motor Integration															
Servo Integration															
Battery Integration															
Avionics Integration															
Testing															
Structural Component Testing															
Electronics Component Testing															
Ground Crew Training															
Flight Test															

7.1 Testing Schedule

7.2 Testing Objectives

The objective of testing was to verify the aircraft's ability to successfully complete all competition missions. This includes verifying the airworthiness and system functionality of the aircraft, completing the required laps within the time limit, demonstrating swift cargo loading while maintaining sufficient payload capacity, and achieving reliable banner deployment and safe landings. Aircraft worthiness testing focused on a few flight regime's, such as cruise, turning, and landing/low speed performance in order to validate the S&C design and confirm that the predicted and actual aircraft behavior and performance matched. Banner testing consisted of evaluating multiple deployment methods, banner sizes, and materials to ensure reliable deployment, while also minimizing mass and drag, utilizing a strain gauge to determine the force of tension from the banner. Cargo testing was performed to determine the optimal number of

ducks and pucks that maximized mission score, without heavily impacting loading time or aircraft performance.

7.3 Subsystem Testing

A series of structural tests and checks will be performed before flight testing the plane: mechanism functionality, balance, and structural integrity. Banner release and payload storage will use smaller test rigs to verify proper functionality before installation into the aircraft. The balance test will consist of lifting at the wing tips to verify CG location. A tilt test ensures all payload remains static even at extreme bank angles. Directly before flight, the aircraft's structural integrity, proper deflection of all control surfaces, and function of banner release mechanisms will be visually checked and verified. Additional testing is described in Table X.

Subsystem Testing			
Sub-System		Test	Objective
Sub-System Tests	Propulsion	Static/Dynamic Thrust Test	Determine and validate thrust data calculated from eCalc and PROM
	Battery	Discharging Test	Determine and validate battery life
	Sensors	Data Collecting Test	Ensure all data from strain gauge and pitot tube are measured accurately
	Release Mechanism	Avionic Test	Ensure mechanism deploys properly
Ground Tests	Control Surfaces	Connection Test, Trim	Make sure all connections are secured and validate control surface trim
	Radio System	Range Test	Make sure all avionics are responsive and move in the correct orientation
	Arming Fuse	Fuse Test	Ensure the fuse can handle a maximum of 100A current
	Sensors	Sensor Test	Power on and confirm readings
	Landing Gear	Load and Drop Test	Validate landing gear strength and ensure hardware security
	Motor Mount	Full Throttle Test	Verify motor and propeller are mounted securely and spin in correct direction
	Full Aircraft	Taxi Test	Verify ground stability and front landing gear trim
Flight Tests	Full Aircraft (M1)	Cruise Test	Confirm stability and control
	Full Aircraft (M1)	Turn Test	Confirm turning performance and bank angle capabilities
	Full Aircraft (M2)	Payload Test	Ensure payload remains secure and test endurance
	Full Aircraft (M3)	Banner Tow Test	Confirm banner deployment and release, test banner drag

7.3.1 Propulsion Testing

A series of propulsion tests were performed with a thrust stand to determine static thrust, current draw, and battery consumption to verify theoretical values. The thrust stand involves a motor mounting system suspended on a set of rails, so that a weight scale can be attached to the back and measure thrust. The testing was done using the Scorpion SII-4025 (330Kv) motor in combination with APC 15x10 propeller. The battery's configuration that was being tested was 8S 3300mAh 100C. The battery was fully charged prior to the testing stage using the charging tool. Before testing, necessary calibration was carried out for the thrust stand by adjusting the position of the slider and making sure the thrust force output by the scale is 0 when the motor is at rest. Then, after the motor wires are connected to the ESC and the main battery, the tester slowly turns the throttle up to 33%, 66%, and 100% while constantly monitoring the thrust and the current data. The performance results are displayed in later section 8.1.1 table (X), where, notably, the current does not surpass 100A. A comparison of predicted and tested values is displayed in Figure X and Y to summarize the propulsion analysis and testing sections.

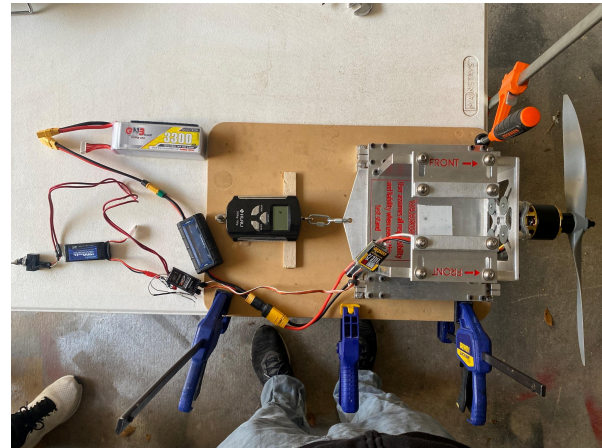


Table X

Propulsion Testing						
Scorpion 4025-330KV, 15x10						
Measurement	Throttle (%)	Thrust (lbs)	Current (A)	Voltage (V)	Power (W)	RPM
Values	33	1.7	3.44	32.7	112.4	2570
	66	7.8	24.3	31.1	756.3	7150
	100	12.2	51.4	29.33	1506.9	8170

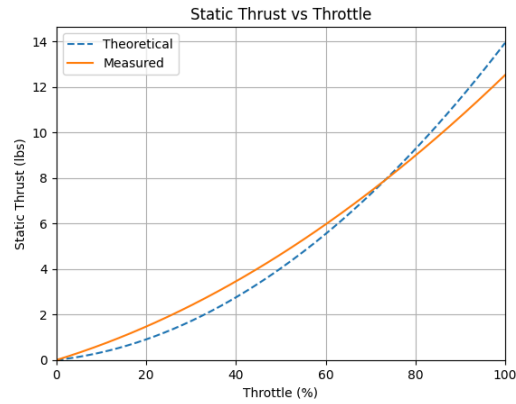


Figure X Static thrust vs throttle

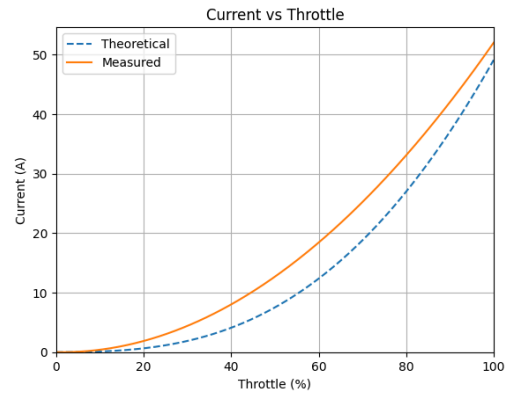
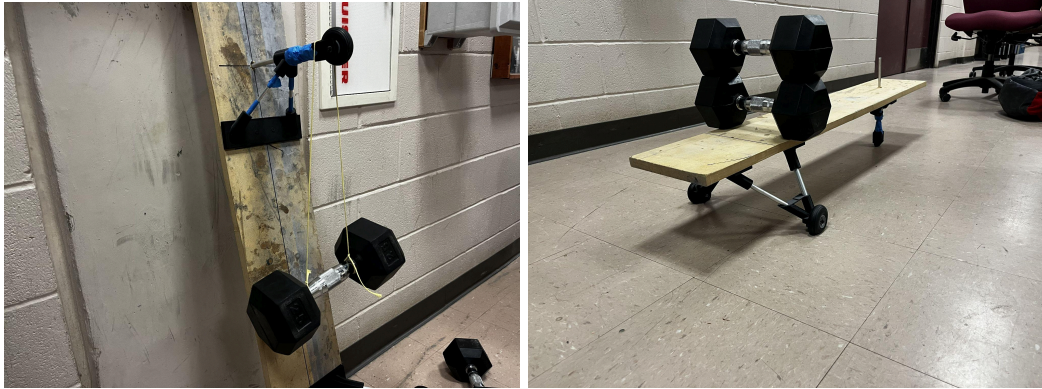


figure Y Current draw vs throttle

The team also conducted an endurance test, where a full flight mission was simulated. The tester first goes up to 100% throttle to initiate the take off condition, then reduces back down to a steady 80% thrust to simulate desired cruise speed. The throttle was also manipulated to mimic the turn and straight level flight for around 4.5 minutes. The temperature of the motor and the ESC were recorded to ensure the system is not overheating. To avoid permanently damaging the battery, it was only drained to 80% of its max capacity, yielding a 4.5 minute flight time, which matches our prediction for 4.9 minute flight window if the battery was to be drained by 85%.

7.3.2. Landing Gear Testing



The design constraints determined that landing gear should be able to withstand 50lbs of static compression, about 2G's, and 25lbs of static shear, or 1G. These static tests were conducted and it passed both with no visible damage, as shown in figure X and Y. Following the compression test a 1ft, 2ft, and 3ft drop test with no additional weight were conducted and passed with no visible damage. The final test was the drop test, as shown in Figure Z. The landing gear structure is able to survive a 3ft drop test with 25lbs placed at the CG, along with surviving a 1ft drop test with 50lbs placed at the CG. The purpose of the drop test is to compare the results of static vs dynamic testing of the landing gear. The idea is under ideal conditions, static forces only, the landing gear should be able to survive, but since the landing gear is being placed on a plane that may land in a handful of different conditions, dynamic testing is necessary. The result was that on our first flight test the plane endured a hard landing due to unstable flight conditions, however the landing gear sustained zero damage and was able to withstand the worst possible conditions.



7.3.3 Banner Testing

It was determined that it was necessary to test the drag characteristics of the banners that the team was planning to use. While banner drag data already existed from commercial applications as well as previous DBF competitions, the data between these sources varied widely, meaning that assumptions made would have to be extremely conservative unless the team were to test their own banners. The team decided to take a similar approach to USC's 2020 DBF team, where a banner was attached to a strain gauge and towed from a truck. The team noted that interference from the truck would have a non negligible impact on the drag of the banner, so the team opted to use a long birch dowel holding the banner through the a sunroof to avoid the more turbulent flow present in the bed of the truck where USC had tested their banner. It is important to note that this test still has a slight amount of error due to the boundary layer on the top of the truck. The test configuration to measure the banner drag is depicted in Fig. XX.



Fig. XX Banner Testing Configuration

Three banners were tested at velocities ranging from 30 mph to 60 mph, with drag measurements taken on 5 mph increments. The three banner lengths were 36 in, 54 in, and 67 in, each with an aspect ratio of 5. The collected drag and velocity data has been plotted in Fig. YY.

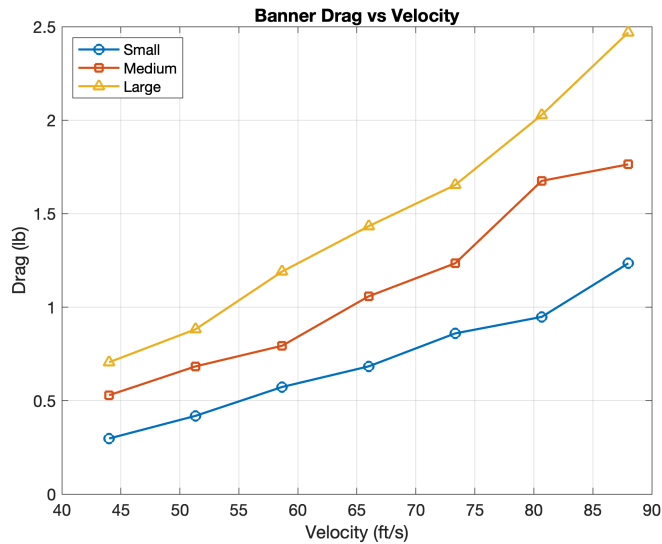


Fig. YY Banner drag vs velocity

The tests showed the expected trend, with larger banners producing more drag, and the banner drag increasing with velocity. This banner drag was then further analyzed with respect to Reynolds number and with the coefficient of drag. The goal of this analysis was to determine what kind of banner would be most efficient, as banners with a lower C_D value would provide a higher score for a lower increase in drag force. The resulting banner drag coefficients were plotted against corresponding Reynolds Numbers in Fig. ZZ.

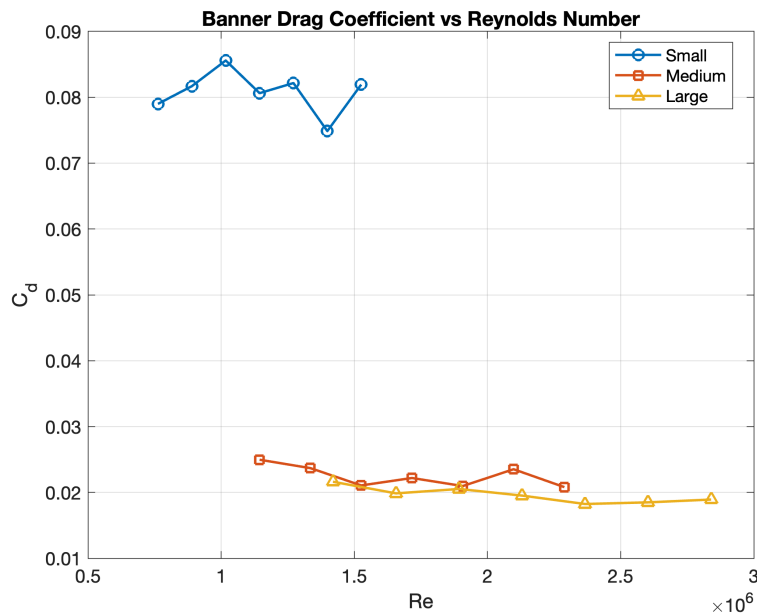


Fig. ZZ Banner Coefficient of Drag vs Reynolds Number

The result shows that the smaller banner consistently had a higher drag coefficient compared to the medium and larger banners, meaning that larger banners would be more efficient than smaller banners. This matches the conclusions drawn by USC's 2020 Banner tests, however, the drag coefficients found

during TAMU's tests were significantly lower than those found by USC. This difference can be attributed to the fact that USC's banner test had been conducted in the bed of a pickup truck rather than above, increasing the drag due to turbulent flow from the truck bed. The difference is also a result of a different banner material, as the team opted for the more durable nylon ripstop over the easier to pack LDPE.

Additional results of the banner test confirmed that the banner tying method and washer weights allowed the banner to stay vertical during flight. It was also noted that the ends of the banners would be frayed after testing, enforcing the need for hemming of the banner for future tests.

7.3.4 Wind Tunnel Testing

TAMU DBF was unable to get access to the low speed wind tunnels until February 2026. The wind tunnel used was a TECQUIPMENT 305 mm Subsonic Wind Tunnel. This tunnel has a test section that is 1 ft wide x 1 ft tall x 2 ft long. To avoid interference from the wind tunnel walls, the aircraft model was scaled down to 13.6% size to have an 8 inch wing span. The model was created with a hole in the rear portion of the fuselage so that the test bar could hold and rotate the plane. The wind tunnel test configuration is shown in fig. AAA.

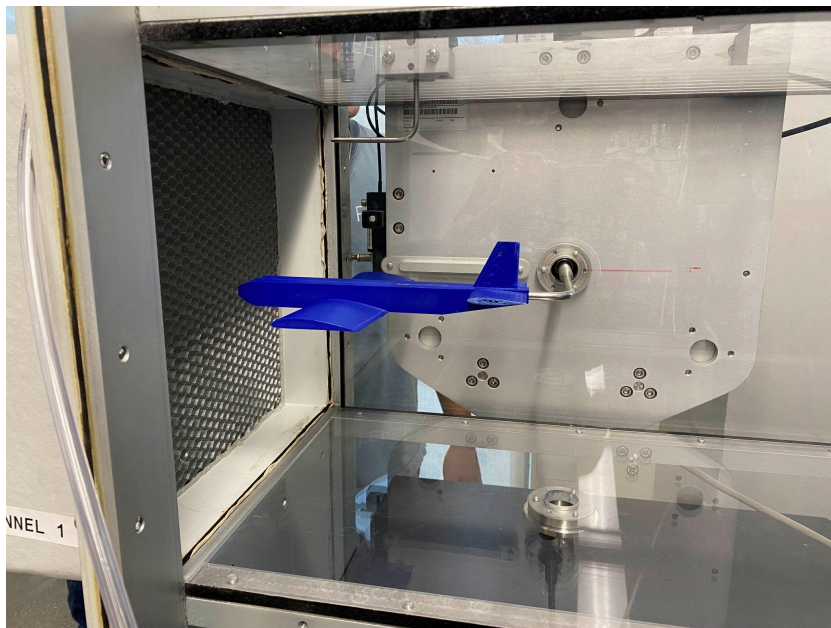


Fig. AAA Wind tunnel model mounted in test section

Since this model is so much smaller than the actual aircraft, matching the Reynold's numbers from flight and in the wind tunnel would require an unrealistically high velocity for a low speed wind tunnel. For this reason, the flow properties seen by the wind tunnel model are not representative of the actual aircraft.

The significant decrease in the model size also suggested that the results obtained from the wind tunnel are not entirely accurate. The results seen in fig. BBB shows a general upward trend in lift.

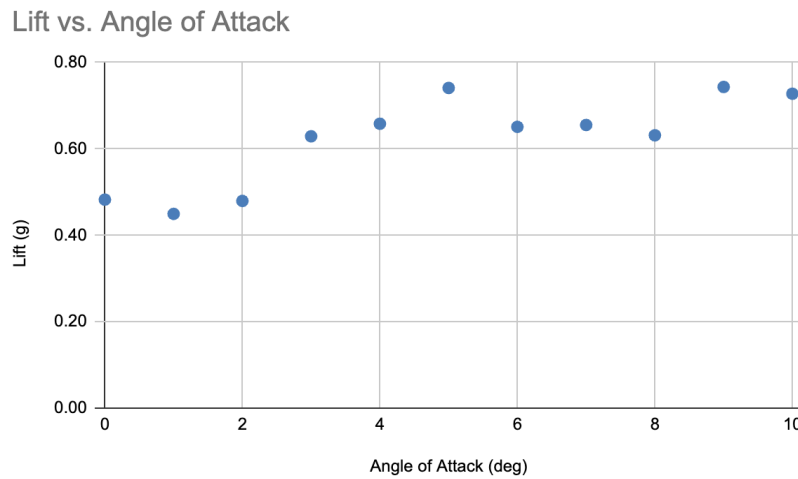


Fig. BBB Lift data at various angles of attack from wind tunnel

7.5 Flight Tests

On 12/6/2025, Flight Test 1 of Mk.1 was conducted to a moderate success. As mentioned in 3.4.1, the aircraft had poor dynamic characteristics which had not shown themselves during the designing phase. Flight Test 1 disproved the original design, and confirmed that the swept wings have poor dynamic characteristics. Although the dynamic characteristics proved to be a major problem in MK1's flight, at cruise speed the aircraft remained stable. However, the low speed characteristics were not acceptable. The aircraft completed several laps with a successful landing, although landing gear damage occurred as the front wheel locked up perpendicular to the landing direction, causing it to skid. Ultimately, while Flight Test 1 demonstrated the aircraft's basic airworthiness, the data gathered from these stability issues and mechanical failures were instrumental in informing the design refinements for MK2.

7.6 Flight Checklists

The TAMU DBF team utilized the following checklists and documentation to ensure safe and repeatable operation of our aircrafts. These checklists consist of a pre-flight inspection, post flight inspection, along with flight test cards in order to create an easy to understand flight profile, all of which can be seen below in **Table X**, **Table Y** and **Table Z**. These documents establish a simple and easy to follow protocol that keep both people and our aircraft safe

Date: ____ / ____ / ____ Aircraft(s): _____			Total Aircraft weight (lb): _____		
Pre-Flight					
Area	☐	FOD Check	Avionics and set up	☐	Plug in arming plug (Aircraft disarmed)
	☐	Establish a clear Flight Line		☐	Propulsion and Receiver battery voltage confirmed Propulsion: ____ volts Receiver: ____ volts
	☐	Set up base camp		☐	Propulsion battery installed
	☐	Establish location of emergency equipment		☐	Receiver battery installed
Fuselage	☐	Perform a visual inspection of all areas		☐	Servos, ESC and receiver are secure
	☐	Aircraft skin tear-free		☐	Servo wires are in correct channels in the receiver
	☐	Ensure wings are secure		☐	Activate flight data recorder and establish base location
	☐	All surfaces and hatches secure		☐	Ensure pitot tube is clear
	☐	Servos linkages and control horns secure		☐	Ensure Strain Gauge is mounted properly and aligned with its strain direction
	☐	Landing gear is secure		☐	Turn on Transmitter
Motors	☐	Motor and Firewall secure		☐	Verify correct CG (<i>Label location in "Final Checks"</i>)
	☐	Ensure propeller is damage-free		☐	Conduct Wingtip Test
Control	☐	Surface deflections are correct and calibrated		☐	Plug in all batteries
	☐	Install propeller		☐	Failsafe checked
	☐	Ensure propeller rotation direction is correct		Final Checks	☐
Preflight Notes		☐			CG Location: _____
<i>Leave notes on how the pre-flight went, and potential improvements that could be made</i>		☐	Flight plan briefed to all personnel		
		☐	Range check		
		☐	Throttle down/safe		
		☐	Remove arming plug (Aircraft armed)		
		☐	Motor run-up test		
		☐	Ground handlers clear of aircraft		
		☐	Confirm take-off direction based on wind		
		☐	Confirm timers are ready		
		☐	Confirm pilot is ready for takeoff		
☐	Ensure all personnel are aware that take off is commencing				

figure X

Post Flight			
Power Down	☐	Throttle down and safe	Debrief Notes <i>Leave notes on how the flight went, and potential improvements that could be made</i>
	☐	Plug in arming plug (Aircraft disarmed)	
	☐	Stop timers	
	☐	Batteries removed	
	☐	Turn off transmitter	
Inspection, Data and Debrief	☐	Perform a visual inspection of aircraft and check airframe health	
	☐	Propulsion system health	
	☐	Data logged	
	☐	Debrief	
	☐	Propulsion and Receiver battery voltages Propulsion: ____ volts Receiver: ____ volts	

figure Y

Flight Test Card Example		
___ / ___ / 2026		
Total Weight (lb)		
CG Locations	Expected/Simulated	Actual
Description of Mission	Example description here. This is an example of a simple maiden flight with 2 laps.	
Battery		
Battery Voltages Start (Fill out on Checklist too)	Propulsion: Avionics:	
Battery Voltages Start (Fill out on Checklist too)	Propulsion: Avionics:	
Times (record times as events happen)		
Mission Start	_____ seconds	
Avionics Start	_____ seconds	
Throttle Up Time	_____ seconds	
Lap Times	Lap 1: _____ seconds Lap 2: _____ seconds	
Landing Time	_____ seconds	
Power Off Time	_____ seconds	
Total Flight Time	_____ seconds	
Procedure		
1. Throttle up and record time 2. Take off 3. 2 laps a. Record each lap time 4. Land and record time		
Notes		

figure Z

8 Performance Results

8.1 Subsystem Performance

8.1.1 Aerodynamics Performance

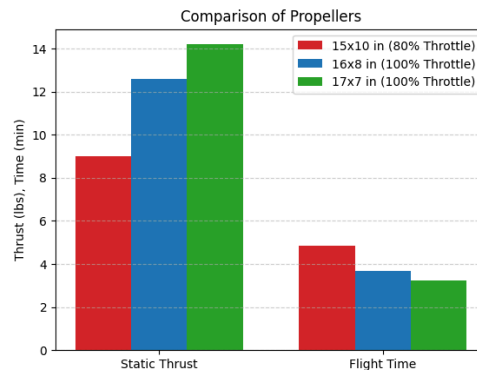
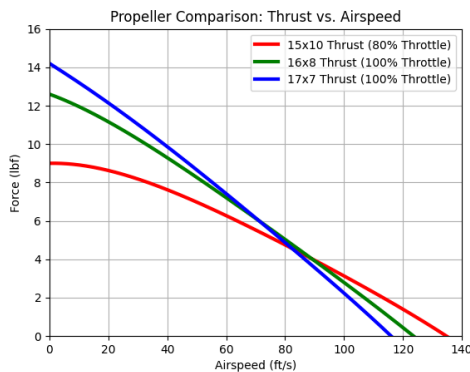
Validation of the aerodynamic performance of the aircraft is critical during the flight test phase. The team chose to record flight test data using an IMU, pitot tube, and accelerometer to record position, velocity, and orientation of the aircraft during flight. The handling characteristics for the MK1 aircraft can be seen in table UU.

Table UU. Flight Data for MK1 flight

Flight Data for MK1 Flight	
Flight Component	Average Velocity (ft/s)
Takeoff	46
Straight	75
Landing	50

The result from the MK1 test flight showed a significantly reduced velocity in the straight section compared to what was expected. Additionally, it was observed that the MK1 aircraft had severe issues with dutch roll dynamic stability, as dutch roll behavior was observed during the landing flight phase. After the flight test, it was determined that a combination of high sweep and inaccurate estimates of aircraft inertia caused the dutch roll behavior. For these reasons, inertia estimates for the MK2 aircraft were much more accurate, and used point masses in XFLR5 rather than relying solely on estimates from the shell CAD model.

8.1.2 Propulsion Performance



8.2 Complete Aircraft Performance

Rev Force 2 consisted of two total iterations: MK1 and MK2. As discussed in previous sections, the MK1 design was found to need significant alterations due to undesirable dutch roll dynamic characteristics. Additionally, it was found that the MK1 design was significantly heavier than predicted. The predicted weight for the MK2 aircraft was 13 lbs, while the actual weight was 17 lbs. This discrepancy also caused reduced performance for the MK1 aircraft. As a result of this testing, the team chose to move to a new MK2 aircraft that solved all of the issues of MK1's performance. The wing sweep was removed, the component weights were more accurately estimated, and the method for including aircraft inertias for the aircraft's dynamic stability was also significantly improved within XFLR5. The result of these improvements was the MK2 aircraft, which has been presented throughout this report. When sized, this

aircraft was expected to lift 6 pucks and 42 ducks, and tow a banner of 36 inches in length. It was also designed to fly for 5 laps in mission 2, and 4 laps in mission 3. After completing the testing plan, it was determined that the aircraft could hold 1 less row of ducks than predicted (39 ducks). The banner test also revealed that the drag estimate used during initial sizing was very conservative, and that the plane would actually be able to tow a 70 inch banner. This, coupled with the new motor selection, enabled an improved performance for the MK2 aircraft.

9 Bibliography

10 Acronyms and more

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\begin{figure}[H]
\centering
\includegraphics[width=0.5\linewidth]{Figures/Wingtip_Vortex.png}
\caption{Wingtip Vortex Footprint}
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\captionof{figure}{Airfoil Lift Comparison}
\label{fig:Foil_Lift}
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\captionof{figure}{Airfoil Drag Comparison}  
\label{fig:Foil_Drag}  
\end{minipage}  
\end{figure}
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